

# The Geometry of Motion

Volume III: Biomechanics

From Rigid Bodies to Biological Systems





# Preface

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**Draft Notice.** This volume is under active development. Chapters 1–3 contain substantial content; Chapters 4–10 are structural outlines with preliminary material that will be expanded in future editions. Chapter 1 includes worked examples and verification exercises. Their numerical consistency does not establish empirical validation of the volume’s biological models.

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The preceding volumes develop geometric and control-theoretic tools using declared mechanical abstractions. This volume asks which abstractions support questions about biological movement. Rigid segments and idealized joints can remain useful; muscle states, tendon compliance, contact, or tissue deformation must be added when the intended output requires them.

Compliance enters constitutive forces and can require additional coordinates; it is not itself an inertial contribution to the mass matrix. Selecting neural excitation as the input also changes the state equations and the meaning of zero-input drift. Chapter 1 establishes these distinctions and shows why a musculoskeletal model is not automatically smooth and control-affine in excitation.

The aim is mathematics constrained by physical evidence. Validation remains specific to the model, population, task, and quantity being predicted. A match to observed motion does not validate every inferred muscle force or tissue load.

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# Nomenclature

## General Conventions

- Scalars are lowercase or Greek letters (e.g.,  $m, t, \theta$ ).
- Vectors are bold lowercase (e.g.,  $\mathbf{x}, \mathbf{u}, \mathbf{q}$ ).
- Matrices are bold uppercase (e.g.,  $\mathbf{A}, \mathbf{B}, \mathbf{M}, \mathbf{J}$ ).
- Expected values and sets are denoted by blackboard bold or script (e.g.,  $\mathbb{E}, \mathcal{S}$ ).

## State and Kinematics

$\mathbf{q} \in \mathbb{R}^n$  Joint configuration vector (generalized coordinates).

$\dot{\mathbf{q}} \in \mathbb{R}^n$  Joint velocity vector (generalized velocities).

$\ddot{\mathbf{q}} \in \mathbb{R}^n$  Joint acceleration vector.

$\mathbf{x} \in \mathbb{R}^{n_x}$  System state vector; commonly  $\mathbf{x} = [\mathbf{q}^T, \dot{\mathbf{q}}^T]^T$  for mechanical systems.

$\mathbf{u} \in \mathbb{R}^m$  Control input vector (e.g., joint torques, muscle activations).

$\boldsymbol{\tau} \in \mathbb{R}^n$  Generalized forces/torques acting on the joints.

$t \in \mathbb{R}$  Time.

## Inertia and Dynamics

$\mathbf{M}(\mathbf{q})$  Mass (inertia) matrix, symmetric positive definite.

$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$  Coriolis and centrifugal force matrix.

$\mathbf{g}(\mathbf{q})$  Gravity vector.

$\mathbf{J}(\mathbf{q})$  Jacobian matrix relating joint velocities to task-space velocities ( $\dot{\mathbf{p}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}}$ ).

## Control and Optimization

$\mathbf{A}_k = \frac{\partial f}{\partial \mathbf{x}}$  Local Jacobian matrix of the state dynamics.

$\mathbf{B}_k = \frac{\partial f}{\partial \mathbf{u}}$  Local Jacobian matrix of the control input.

$V(\mathbf{x})$  or  $\mathcal{V}$  Value function or Lyapunov function.

$\mathbf{Q}$  State penalty matrix in LQR/DDP.

**R** Control penalty matrix in LQR/DDP.

**K** Feedback gain matrix ( $\delta \mathbf{u} = -\mathbf{K}\delta \mathbf{x}$ ).

$\gamma$  A trajectory or flow, mapping time and initial conditions to states.

## Affine Control and Counterfactuals

$f(\mathbf{x})$  Drift vector field: complete autonomous evolution of the declared effective plant when  $\mathbf{u} = \mathbf{0}$ .

$G(\mathbf{x})$  Input (control) matrix field mapping control inputs to state derivatives.

$\dot{\mathbf{x}} = f(\mathbf{x}) + G(\mathbf{x})\mathbf{u}$  Control-affine system equations of motion.

$\mathbf{x}^{\text{ZTCF}}(t)$  Forward Zero-Torque Counterfactual trajectory under  $\mathbf{u}(t) \equiv \mathbf{0}$ .

$\mathbf{a}_{\text{ZVCF}}(\mathbf{q}, \mathbf{z})$  Instantaneous Zero-Velocity Counterfactual acceleration with velocity and declared control set to zero.

$\text{DCR}_{W, \mathcal{U}}(\mathbf{x})$  Drift-Control Ratio in a declared acceleration or task-projected space:  $\frac{\|W\mathbf{a}_{\text{drift}}\|_2}{\sup_{\mathbf{u} \in \mathcal{U}} \|WB_a\mathbf{u}\|_2 + \varepsilon}$ .

## Biomechanics and Motor Control

$a_i \in [0, 1]$  Muscle activation level.

$\ell_i^M$  Muscle fiber length.

$v_i^M$  Muscle fiber contraction velocity.

$\ell_i^{MT}$  Musculotendon unit length.

$\mathbf{F}_{\text{muscle}}$  Force generated by muscles.

$\mathbf{W}$  Muscle synergy matrix (weights).

$\mathbf{c}(t)$  Muscle synergy activation vector over time.

## Geometry and Manifolds

$SE(3)$  Special Euclidean group of rigid body motions.

$SO(3)$  Special Orthogonal group of 3D rotations.

$\mathfrak{se}(3)$  Lie algebra of  $SE(3)$ , space of spatial twists (velocities).

$\mathcal{M}$  A smooth manifold.

$T_{\mathbf{x}}\mathcal{M}$  Tangent space at point  $\mathbf{x}$ .

# Chapter 1

## How Biology Differs From Engineering

The golf swing connects neural excitation, muscle and tendon dynamics, joint motion, ground contact, and club deformation. A model can represent selected parts of this chain accurately while leaving others unresolved. Reproducing clubhead motion does not, by itself, establish which muscles generated it; reproducing net joint moments does not establish local tissue stress. This chapter develops the assumptions needed for a particular question and the corresponding tests of those assumptions.

The mathematical tools of the preceding volumes remain useful. Their application depends on the chosen state, input, constraints, and output. Biology changes those choices; it does not suspend mechanics. Numerical examples in this chapter are manufactured checks of equations, not measured human parameters or recommendations for a golfer.

### 1.1 The Rigid-Body Assumption and Its Limits

A rigid body has a fixed mass distribution in its own frame. A revolute joint permits one relative rotation and constrains the other five relative motions. Both are idealizations useful in biological and engineered systems. Bones deform, but their deformation need not be resolved for every movement question. Engineered shafts, transmissions, and sensors can also have compliance, backlash, noise, delay, and temperature dependence. The relevant distinction is between a declared approximation and the behavior it must explain.

For a golf model, declare the segment inventory, coordinate frames, joint constraints, and external contacts before selecting actuators. A shoulder rotation model may suffice for an endpoint-kinematics question while being inadequate for joint-contact loading. A hand-club constraint can transmit forces and moments even when its constrained coordinates are not actuated. Calling an anatomical joint “six degree of freedom” also requires care: relative pose has six components, but constraints and tissue mechanics may couple or strongly resist some motions. Coordinate count is not a count of independently controllable directions.

#### Biology vs. Engineering 1.1. Modeling Assumptions to Test

##### **Structure**

Choose rigid segments, compliant modes, or a continuum description according to the deformation that affects the target output.

##### **Actuation**

Specify actuator states and force, velocity, power, delay, and saturation limits where they matter.

**Sensing**

Identify observed quantities, error, sampling, filtering, and latency. A reconstructed state is not itself a sensor reading.

**Adaptation**

State the time horizon over which parameters are fixed. Fatigue, learning, and material changes require additional hypotheses.

**Redundancy**

Establish the rank of the relevant map and its feasible domain before claiming alternative solutions or compensation.

Motor torque can depend on speed, current dynamics, and thermal limits; muscle force can depend on activation, fibre length, velocity, and history. An ideal torque source deliberately suppresses these dependencies. It can be a useful first model of either actuator, provided conclusions remain about net loads and motion. A richer biological model should earn its added complexity through improved, independently tested predictions.

## 1.2 Compliance, Timescales, and the Chosen Output

### 1.2.1 Adding Deformation Requires Adding Dynamics

A free rigid club has six pose coordinates locally, not one grip angle. A one-angle club is a constrained planar abstraction. Flexible modes add coordinates to the selected rigid model. Write  $\mathbf{z} = (\mathbf{q}, \boldsymbol{\eta})$ , where  $\mathbf{q} \in \mathbb{R}^n$  describes the rigid coordinates and  $\boldsymbol{\eta} \in \mathbb{R}^r$  the retained deformation amplitudes. The selector  $E = [0_{r \times n} \ I_r]$  gives  $E\mathbf{z} = \boldsymbol{\eta}$ . In a local chart with  $\dot{\mathbf{z}}$  as its generalized velocity, an illustrative model is

$$M(\mathbf{z})\ddot{\mathbf{z}} + h(\mathbf{z}, \dot{\mathbf{z}}) + E^T(K\boldsymbol{\eta} + D\dot{\boldsymbol{\eta}}) = B(\mathbf{z})\boldsymbol{\tau} + J_c(\mathbf{z})^T\boldsymbol{\lambda}. \quad (1.1)$$

All terms are generalized forces in the same augmented coordinate space.  $M$  includes rigid–flexible inertial coupling;  $h$  contains remaining inertial and gravitational terms. Constant  $K, D$  describe a local linear elastic and viscous approximation, not arbitrary tissue behavior.  $B$  maps actuator loads  $\boldsymbol{\tau}$  into the augmented coordinates, while  $J_c$  is the constraint Jacobian and  $\boldsymbol{\lambda}$  its conjugate reaction loads. Contact equations complete the model. Adding an unprojected modal spring force to a rigid-body equation would omit both dimensions and coupling.

The shaft’s relevance depends on the output: small deformation can matter for club-face orientation while contributing little to whole-body motion. Ground contact and grip loads determine modal excitation; controller bandwidth alone does not describe that loading spectrum.

### 1.2.2 A Frequency Comparison With a Checkable Meaning

For a linear mode, let

$$m\ddot{\eta} + d\dot{\eta} + k\eta = F(t), \quad \omega_n = \sqrt{k/m}, \quad \zeta = \frac{d}{2\sqrt{km}}.$$

For a harmonic load of angular frequency  $\Omega$ , steady-state displacement per unit force is

$$H(\Omega) = \frac{1}{k - m\Omega^2 + i d \Omega}, \quad kH(\Omega) = \frac{1}{1 - (\Omega/\omega_n)^2 + 2i\zeta\Omega/\omega_n}.$$

At low frequency,  $H \simeq 1/k$ : quasi-static compliance can still produce important deflection. Near resonance, damping and coupling matter. Far above the mode's frequency, inertia dominates the forced response; that does not establish rigidity during a transient with initial deformation and broadband loads.

#### Biomechanical Principle 1.1. The Timescale Separation Principle

Compare the relevant load spectrum and initial transients with modal frequencies and damping, then propagate omitted deformation to the output. Quasi-static reduction requires negligible dynamic error over that spectrum; a rigid approximation additionally requires negligible deformation error for the task. Neither follows from controller bandwidth alone.

For a manufactured translational mode with  $m = 1$  kg,  $k = 100$  N/m, and  $d = 1$  N s/m,  $\omega_n = 10$  rad/s and  $\zeta = 0.05$ . At  $\Omega = 1, 10, 100$  rad/s, the amplitude relative to static compliance,  $|kH|$ , is approximately 1.010, 10, and 0.0101. The high-frequency response is nearly opposite in phase to the force. A constant static gain cannot reproduce all three regimes. These are illustrative parameters, not measurements of a golf shaft.

## 1.3 From Excitation to Joint Power

### 1.3.1 Force Definitions and Units

A common Hill-type idealization separates activation, normalized active force–length and force–velocity curves, and parallel passive force [19]. With positive fibre velocity denoting lengthening, write the illustrative undamped fibre force as

$$F_M = F_0 \left[ a f_L(\ell_M/\ell_0) f_V(\dot{\ell}_M/v_0) + f_{PE}(\ell_M/\ell_0) \right]. \quad (1.2)$$

$\ell_M$  is fibre length,  $\ell_0$  its optimal reference length, and  $a \in [0, 1]$  the activation state.  $F_0$  is the maximum active isometric fibre-force scale at optimal length; the curve values are dimensionless.  $v_0 > 0$  is a declared velocity scale, not a universal human shortening speed. The parallel-elastic term  $f_{PE}$  is scaled by the same outside factor  $F_0$ . Other models add damping or history dependence and use different normalizations.

Fibre force is not automatically tendon force. In a massless, ideal pennated equilibrium model,  $F_T = F_M \cos \alpha$ , where  $\alpha$  is the pennation angle between fibres and tendon. This relation requires a tendon constitutive law and compatible fibre–tendon geometry. A compliant tendon can change length while the whole musculotendon path length changes little. Joint angular velocity therefore does not by itself determine fibre shortening velocity. Model-specific equilibrium, slack, and regularization conditions must be checked before solving for fibre velocity.

### 1.3.2 Moment Arms Must Preserve Power

For differentiable, time-independent musculotendon routing  $\ell_{MT}(\mathbf{q})$ , define

$$R_{ji} = -\frac{\partial \ell_{MT,i}}{\partial q_j}, \quad \boldsymbol{\tau}_M = R(\mathbf{q})\mathbf{F}_T. \quad (1.3)$$

For rotational coordinates,  $R_{ji}$  has length units and the conjugate load is a moment. Translational coordinates instead have force as the conjugate load. Virtual work gives the sign-sensitive check

$$\dot{\ell}_{MT} = -R^T \dot{\mathbf{q}}, \quad \boldsymbol{\tau}_M^T \dot{\mathbf{q}} = -\mathbf{F}_T^T \dot{\ell}_{MT}.$$

A positive tensile force delivers positive skeletal power when its path shortens. Fibre power can differ from path power because tendon energy storage and other internal processes intervene. For constrained coordinates or speeds distinct from  $\dot{\mathbf{q}}$ , apply the velocity map before comparing power.

For a numerical check, take rotational coordinates and

$$R = \begin{bmatrix} 0.03 & -0.03 & 0 \\ 0 & 0.02 & -0.02 \end{bmatrix} \text{ m}, \quad \mathbf{F}_T = \begin{bmatrix} 100 \\ 80 \\ 60 \end{bmatrix} \text{ N}, \quad \dot{\mathbf{q}} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \text{ s}^{-1}.$$

Then  $\boldsymbol{\tau}_M = (0.6, 0.4)^T$  N m and  $\dot{\ell}_{MT} = (-0.06, 0.08, -0.02)^T$  m/s. Both expressions for skeletal power give 0.8 W. A moment-arm sign error can look plausible in a torque plot while failing energy accounting immediately.

#### Biomechanical Principle 1.2. Force-Velocity Asymmetry in Multi-Joint Tasks

Force capacity depends on each muscle's fibre state, activation, parameters, and constitutive curves. A biarticular muscle couples multiple joint motions through its routing and tendon dynamics. Evaluate those states before classifying its contraction or constructing the feasible joint-load set; joint rotation alone does not specify eccentric or concentric fibre contraction.

In a downswing, the same visible wrist motion can be compatible with different tendon strains, activation histories, and antagonist forces. Muscle modeling makes those alternatives calculable under assumptions; it does not make video sufficient to identify which alternative occurred.

## 1.4 Two Different Kinds of Redundancy

Let the task be  $\mathbf{y} = \phi(\mathbf{q})$ , with Jacobian  $J_y = \partial\phi/\partial\mathbf{q}$ . For  $n$  independent local configuration coordinates, infinitesimal motions preserving  $\mathbf{y}$  satisfy  $J_y \delta\mathbf{q} = 0$ , with dimension  $n - \text{rank } J_y$ . A three-joint planar arm controlling a 2D endpoint position has a one-dimensional local nullspace only where the Jacobian has rank two. At singular configurations the rank changes. A finite displacement requires following the nonlinear level set, not extrapolating one tangent direction indefinitely.

Muscle-force redundancy concerns a different map. At fixed geometry, an instantaneous force-allocation model solves  $R\mathbf{F}_T = \boldsymbol{\tau}_{\text{req}}$ , with a linear equality solution space of

dimension  $p - \text{rank } R$  when nonempty, for  $p$  muscle paths. Nonnegative tensile forces and state-dependent capacities restrict this space. In the dimensionless algebraic example

$$R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad \tau_{\text{req}} = \begin{bmatrix} 3 \\ 3 \end{bmatrix},$$

all equality solutions are  $(3 - t, 3 - t, t)^T$ . Bounds  $0 \leq F_i \leq 3$  leave  $0 \leq t \leq 3$ . Other loads or capacities can leave a point or no solution. This is an allocation example, not a human moment-arm model. It shows why a nullspace count cannot guarantee compensation for a fatigued muscle. These candidate forces must still satisfy muscle dynamics: if every internal state is already fixed, a constitutive model may fix tendon force as well. An allocation equality alone does not supply a realizable excitation history.

Even equal nullity does not identify equal physical freedom. The task map  $J_y = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$  has null vector  $(1, -1, 1)^T$ , whereas the example  $R$  has null vector  $(-1, -1, 1)^T$ . Their domains also represent different quantities: configuration changes and muscle forces. Matching hand position does not guarantee matching velocity, joint loading, energy use, or future motion.

The uncontrolled manifold hypothesis tests whether variability is preferentially organized along directions preserving a selected task variable [26]. Its original sit-to-stand analysis supports a task-dependent interpretation of joint variability. It is not a direct measurement of a neural projection operator or proof that all task-null directions are left uncontrolled. Coordinate scaling, task choice, noise, and dynamics influence interpretation. In golf, variability preserving clubhead position at one instant can still change face orientation, impact timing, or the subsequent response to a perturbation.

## 1.5 Choosing the Model by the Question

### 1.5.1 Rigid Segments With Generalized-Load Actuators

This model examines motion, power, momentum, and net loads without resolving individual muscles. Inverse dynamics requires kinematics, inertial parameters, external loads, and constraints; it does not identify individual muscle forces from motion alone. Add actuator bounds when testing trajectory feasibility. An unconstrained optimal torque history is a mathematical result, not evidence that a golfer can execute it.

For clubhead speed, this level can clarify how mass distribution, segment motion, and external forces interact. A sensitivity calculation can ask whether plausible parameter changes or shaft deformation alter the conclusion. If they do, examine the omitted physics before interpreting the result as a mechanism of a human swing.

### 1.5.2 Rigid Segments With Muscle and Tendon States

This adds routing, activation, and constitutive assumptions. A forward simulation requires excitations and initial states. Estimating excitation from observed motion requires further data, a control hypothesis, or an optimization criterion. Multiple solutions may fit the same motion. Report residual forces, reserve actuators, parameter sensitivity, and objective sensitivity rather than presenting one optimum as an observed neural strategy.

OpenSim supports musculoskeletal modeling and simulation; a typical model combines rigid bodies, joints, and force elements [7]. Such a model is not automatically a finite-element

model of deforming bone or soft tissue. Joint reactions are resultant loads conditional on modeled actuators and constraints, not local stress fields.

### 1.5.3 Deformation or Tissue Stress as an Output

Flexible modes, continuum models, or coupled multibody–finite-element models resolve deformation when the question requires it. Local stress requires geometry, constitutive properties, boundary conditions, and loading; numerical resolution and constitutive validation are separate checks. A resultant force and moment constrain integrals of traction without uniquely determining its spatial distribution. A calculated stress field is not an injury probability without a validated injury or failure model.

These choices are not a universal fidelity ranking. Extra parameters can increase uncertainty when measurements cannot identify them. Real-time feasibility depends on model, algorithm, hardware, accuracy, and update rate; a fixed degree-of-freedom threshold cannot select the method. State an output tolerance, compare simpler and richer models against independent observations, and retain the least complexity that supports the claim.

## 1.6 Mass Properties Need Accounting and Frames

### 1.6.1 A Complete Segment Inventory

Anthropometric coefficients require a population, segmentation scheme, landmarks, and parameter conventions. De Leva’s study adjusts landmark references of Zatsiorsky–Seluyanov segment parameters [6]; it is not a generic update of Dempster cadaver coefficients. Mixing incompatible segmentations can double-count or omit mass. Scaling population estimates also does not make them subject-specific measurements.

An inconsistent inventory illustrates the danger: central fractions 0.081 and 0.497 combined with six unilateral fractions 0.0271, 0.0162, 0.0061, 0.1416, 0.0433, and 0.0137. Their one-row sum is 0.826; counting both sides gives 1.074. Neither accounts for one body. Renormalizing conceals the provenance problem. No empirical replacement table is asserted here without verification of its source, landmarks, and segment definitions.

For a synthetic accounting exercise, use central fractions 0.10, 0.40 and six paired fractions (0.05, 0.03, 0.02, 0.09, 0.04, 0.02) per side. Their bilateral sum is one. For 75 kg, the central masses are 7.5, 30 kg and each side has (3.75, 2.25, 1.50, 6.75, 3.00, 1.50) kg. This tests bookkeeping only; it provides neither human lengths nor centres of mass or inertias.

### 1.6.2 Centre of Mass and Spatial Inertia

Let a segment’s local centre of mass be  $\mathbf{c}_i$ , its local-to-world rotation matrix  $Q_i \in SO(3)$ , and its world origin  $\mathbf{p}_i$ . Then

$$\mathbf{r}_i = \mathbf{p}_i + Q_i \mathbf{c}_i, \quad \mathbf{r}_{COM} = \frac{\sum_i m_i \mathbf{r}_i}{\sum_i m_i}.$$

Body mass and height alone cannot specify these world positions. A pose and kinematic model are needed, with separate left and right instances. For masses 2, 1 kg at (0.2, 0, 0), (0.4, 0.15, 0) m, respectively, the centre is (4/15, 0.05, 0) m. Translating both positions translates the centre by the same amount, a useful frame-consistency check.

A scalar  $I = m(\rho L)^2$  describes inertia about a specified axis when  $\rho L$  is its radius of gyration. General 3D dynamics needs a symmetric tensor  $I_C$  about the centre of mass in declared axes. For an angular-first twist  $V = (\boldsymbol{\omega}, \mathbf{v}_O)$ , all expressed in one frame, let  $\mathbf{c}$  point from origin  $O$  to the centre and  $C = [\mathbf{c}]_\times$ . Then

$$\mathcal{I}_O = \begin{bmatrix} I_C - mCC & mC \\ -mC & mI_3 \end{bmatrix}.$$

The signs follow from  $\mathbf{v}_C = \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{c} = \mathbf{v}_O - C\boldsymbol{\omega}$ , giving

$$\frac{1}{2}V^T \mathcal{I}_O V = \frac{1}{2}\boldsymbol{\omega}^T I_C \boldsymbol{\omega} + \frac{1}{2}m\|\mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{c}\|^2.$$

Positive mass and positive-definite  $I_C$  make this form positive definite. Physical inertia tensors also satisfy principal moment triangle inequalities; positive eigenvalues alone are not a complete realizability test. Point or ideal line masses can have singular rotational inertia. Declare which case an implementation accepts.

## 1.7 When the Control-Affine Framework Applies

### 1.7.1 Choose the Input Before Splitting Drift and Control

Torque-driven and excitation-driven models describe different interventions. For a smooth fixed-contact model in independent coordinates, let  $\mathbf{s}$  collect muscle states, with composite state  $\mathbf{x} = (\mathbf{q}, \mathbf{v}, \mathbf{s})$ . Assume  $M$  is invertible on the domain under analysis. A possible description is

$$\dot{\mathbf{q}} = N(\mathbf{q})\mathbf{v}, \quad M(\mathbf{q})\dot{\mathbf{v}} = \mathbf{f}_{\text{mech}}(\mathbf{q}, \mathbf{v}, \mathbf{s}), \quad \dot{\mathbf{s}} = h_s(\mathbf{q}, \mathbf{v}, \mathbf{s}, \mathbf{u}).$$

$N$  accounts for representations whose coordinate derivatives differ from generalized speeds. Reactions must be solved consistently or eliminated on the constraint manifold. Here excitation acts through muscle states, with no separate direct excitation term in mechanics. Input affinity depends on  $h_s$  and any other input-dependent equations.

#### Biomechanical Principle 1.3. The Extended Affine Structure

A declared model is control-affine in its chosen input if

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{G}(\mathbf{x})\mathbf{u}. \quad (1.4)$$

Nonlinear state dependence in  $\mathbf{f}, \mathbf{G}$  is permitted. For example,  $\dot{a} = (u - a)/T$  with constant  $T > 0$  is affine in excitation. If remaining muscle-state equations depend on activation and mechanics but not directly on excitation, their nonlinearity does not invalidate this structure. Establish these assumptions for the selected model; do not infer them from the word “musculoskeletal.”

Consider instead the branch-dependent law documented for OpenSim’s [first-order activation model](#):

$$\dot{a} = \frac{u - a}{T(u, a)}, \quad T(u, a) = \begin{cases} T_{\text{act}}(0.5 + 1.5a), & u > a, \\ T_{\text{deact}}/(0.5 + 1.5a), & u \leq a. \end{cases}$$

For a manufactured evaluation without an activation floor, choose  $a = 0.5$ ,  $T_{act} = 0.01$  s, and  $T_{deact} = 0.04$  s. The rates at  $u = 0, 0.5, 1$  are  $-15.625, 0, 40$  s<sup>-1</sup>. An affine input map would give the midpoint rate  $(-15.625 + 40)/2 = 12.1875$  s<sup>-1</sup>, not zero. Each branch is affine in  $u$  for fixed state, but the full map is not. The rate is continuous at  $u = a$ , with generally unequal one-sided slopes. Activation floors and clamping in an implementation need their own treatment; this arithmetic does not reproduce every software variant.

#### Remark 1.1. Dimensional Expansion of the State Space

For  $n$  independent coordinates,  $p$  activations, and  $p$  independent fibre length states, a local state dimension is  $2n + 2p$ . A rigid-tendon model may omit fibre-length states; fatigue or history states increase the count. Algebraic constraints and redundant pose representations change the distinction between stored variables and independent states. Derive dimension from the equations rather than a universal number of states per muscle.

Lie-bracket calculations require appropriate smooth vector fields and a specified admissible input set. Contraction arguments also require a suitable metric, invariant domain, and the theorem's differential conditions. Input bounds, contact transitions, and nonsmooth force laws can limit conclusions. Smooth analysis within a contact mode is not a proof across impacts or changes in contact. Dimensionality is a computational challenge; validity of the mathematical hypotheses comes first.

Zero excitation does not instantly erase activation or tendon strain. With constant  $T$ , activation subsequently decays as  $a(t) = a(0)e^{-t/T}$  while other states evolve. Zero-input drift can thus include force associated with prior excitation. It is not synonymous with anatomically passive force, zero joint torque, or zero power. A golf counterfactual must specify what is removed and which initial states, contacts, and feedback rules are retained.

## 1.8 Connecting the Parts of the Swing Argument

The choices form a chain of inference. Kinematics and external loads, with mass properties and contacts, constrain net dynamics. Routing and constitutive assumptions map candidate muscle states into those loads. Activation dynamics connect states to candidate excitations. Compliance changes both transmission and the output being controlled. Redundancy leaves alternatives that an objective or further measurement may restrict. Every added explanation introduces a corresponding test.

A claim that a golfer increases clubhead speed through a particular muscle action requires more than a faster clubhead trajectory. Test whether the conclusion persists across plausible inertial parameters, grip constraints, tendon properties, allocation objectives, and shaft models. Separate an instantaneous power identity from a finite-time causal intervention: removing one torque or excitation changes later states and may change loads supplied by feedback. Report the intervention and its uncertainty alongside the predicted effect.

The resulting explanation should distinguish measured quantities, mechanical calculations, biologically conditioned inferences, and hypotheses about control or coaching. That distinction lets a simple model provide strong insight while identifying what remains unresolved about the swing.

## Exercises With Verification Targets

1. **Mass Accounting.** Reproduce both erroneous inventory sums and the synthetic 75-kg inventory. Explain why a unit sum cannot validate anatomical provenance or supply missing centre-of-mass locations.
2. **Compliance.** Reproduce the three values of  $|kH|$  above. A hypothetical 5-Hz mode completes  $5(0.25) = 1.25$  cycles in 0.25 s. Explain why this count supplies neither excitation amplitude nor output error and cannot decide whether shaft deformation is negligible.
3. **Power.** Verify the 0.8-W example by both routes. Identify the energy-storage terms needed to equate skeletal and contractile-element power for a compliant tendon.
4. **Redundancy.** Derive both null vectors and the interval  $0 \leq t \leq 3$ . For three joints and six muscle paths, state the ranks needed for task nullity one and force nullity three. Explain why neither number counts identifiable neural excitation histories.
5. **Mass Frames.** Reproduce the two-mass centre of mass and its translation covariance. With upward  $y$  and gravity magnitude  $g$ , compute  $U = \sum_i m_i g y_i = 0.15g$ . State the reference level and why mass fractions alone cannot determine potential energy.
6. **Spatial Inertia.** Use  $m = 2$  kg and

$$\mathbf{c} = (0.1, -0.2, 0.3) \text{ m}, \quad I_C = \text{diag}(0.02, 0.03, 0.04) \text{ kg m}^2.$$

Check symmetry, positivity, principal moment inequalities, and both energy expressions for  $\boldsymbol{\omega} = (1.5, -2, 0.5) \text{ s}^{-1}$  and  $\mathbf{v}_O = (-1.2, 0.8, -0.4) \text{ m/s}$ . Nonzero translation tests the cross-block signs.

7. **Input Affinity.** Compare the branch-dependent example with the constant- $T$  model,  $T = 0.02$  s, giving rates  $(-25, 0, 25) \text{ s}^{-1}$  at the same inputs and activation. Explain why nonlinear activation dependence alone does not disprove input affinity, and why zero excitation is not an instantaneous relaxed state.
8. **Model Selection.** Specify separate evidence requirements for clubface pose, net wrist moment, individual tendon force, and local stress. For a force-velocity calculation, identify the missing curve, normalization, fibre state, activation, and tendon assumptions before assigning an eccentric-to-isometric force ratio.

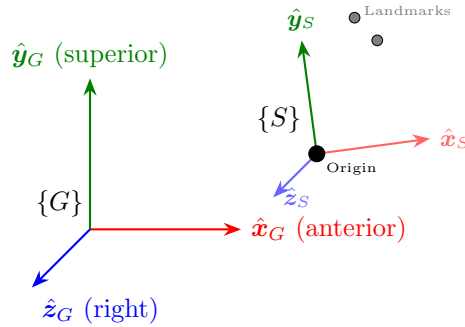
## Chapter 2

# Musculoskeletal Modeling Conventions

The same measured motion can yield different reported angles when frame axes, calibration poses or rotation conventions differ. A useful model must connect those choices to its inertial parameters and force calculations. Consistency of coordinates is necessary for physical interpretation; it does not establish the accuracy of the measurements or identify the subject's controller.

### 2.1 Anatomical Reference Frames

Biomechanical analysis requires a standardized language for describing body position and motion. The International Society of Biomechanics (ISB) has published recommendations for joint coordinate systems [31, 32]. These are joint-specific recommendations, not a single universal axis assignment or angle sequence.



#### 2.1.1 The Anatomical Position

The **anatomical position** provides a reference for directional terminology: the subject stands upright with arms at the sides and palms facing forward. It does not automatically define every model's zero joint configuration. Record the calibration pose, anatomical landmarks, segment-frame offsets and the convention used to report zero. A subject's measured standing pose need not coincide with an idealized anatomical neutral pose.

**Definition 2.1. Global Reference Frame**

For the examples here, choose a fixed global (laboratory) frame  $\{G\}$  as:

- $\hat{x}_G$ : anterior (forward)
- $\hat{y}_G$ : superior (upward)
- $\hat{z}_G$ : right

This is a right-handed coordinate system, aligned with the subject in the chosen reference pose; it does not rotate with the subject. Right is not synonymous with lateral on both limbs. Other laboratory and software systems use different conventions. **Verify handedness, axis directions, units and frame origins before combining data.**

**2.1.2 Segment Reference Frames**

In a rigid-segment model, a local frame is attached to each segment. Anatomical frames can be constructed from landmarks and estimated joint centers; a marker cluster's technical frame generally requires a separate anatomical calibration. Skin-mounted markers need not remain rigidly attached to the underlying bone. Report the landmark definitions and estimation method, rather than assuming that a measured technical frame is already the desired anatomical frame.

**Example 2.1. The Thigh Reference Frame**

Wu et al.'s femoral frame for hip reporting [31], Section 4.4, uses the following definitions:

- **Origin:** hip joint center
- $\hat{y}_T$ : from the epicondyle midpoint toward the hip center
- $\hat{z}_T$ : in the plane formed by the two epicondyles and the hip center, perpendicular to  $\hat{y}_T$ , pointing right
- $\hat{x}_T$ : cross product  $\hat{y}_T \times \hat{z}_T$  (anterior direction)

Normalize the axes and check that the landmarks define a nondegenerate plane. A frame located at the knee can also be useful, but its origin must be transported explicitly; it is not this attributed ISB origin.

The recommendation favors functional hip-center estimation when adequate relative motion is available and discusses predictive alternatives. Anthropometric regression, such as Harrington et al.'s MRI-based method [13], estimates a center; it does not make that center an exact observed landmark or remove calibration and population uncertainty.

### 2.1.3 Connection to Screw Theory

In the language of Volume 0, each segment frame  $\{S_i\}$  is related to the global frame  $\{G\}$  by a homogeneous transformation matrix  $T_{GS_i} \in \text{SE}(3)$ :

$$T_{GS_i} = \begin{bmatrix} R_{GS_i} & \mathbf{p}_{GS_i} \\ \mathbf{0}^T & 1 \end{bmatrix}, \quad (2.1)$$

where  $R_{GS_i} \in \text{SO}(3)$  maps segment-coordinate vectors into global coordinates; its columns are the segment axes expressed globally. The vector  $\mathbf{p}_{GS_i} \in \mathbb{R}^3$  locates the segment origin globally. Thus a point satisfies  $\mathbf{x}_G = R_{GS_i} \mathbf{x}_{S_i} + \mathbf{p}_{GS_i}$ .

The *relative configuration* of segment  $j$  with respect to proximal segment  $i$  is:

$$T_{S_i S_j} = T_{GS_i}^{-1} T_{GS_j}. \quad (2.2)$$

The rotation block is  $R_{S_i S_j} = R_{GS_i}^T R_{GS_j}$ ; the translation is  $R_{GS_i}^T (\mathbf{p}_{GS_j} - \mathbf{p}_{GS_i})$ . A common rigid change of laboratory coordinates leaves this relative transform unchanged. Anatomical recalibration is different: changing the attached segment frames changes the reported relative coordinates and must also change any frame-dependent inertial data.

## 2.2 Joint Coordinate Systems

Joint-coordinate conventions relate relative orientation to anatomical terms. The chosen axes, positive directions, limb side, neutral offsets and angle branches must all be stated. An Euler/Cardan decomposition is a coordinate description of the resulting orientation, not a measured temporal sequence of three isolated motions.

### 2.2.1 The Grood–Suntay Convention for the Knee

The knee joint coordinate system, originally proposed by Grood and Suntay (1983) [11], decomposes knee motion into:

1. **Flexion-Extension** ( $\alpha$ ): rotation about the femoral medio-lateral axis (the “fixed” axis of the femur)
2. **Abduction-Adduction** ( $\beta$ ): rotation about the “floating” axis (perpendicular to both the femoral and tibial body-fixed axes)
3. **Internal-External Rotation** ( $\gamma$ ): rotation about the tibial longitudinal axis (the “body-fixed” axis of the tibia)

The floating axis is perpendicular to the two body-fixed axes; its directed cross product depends on the declared sign convention. It is undefined when those axes are parallel. With suitable segment-axis assignments, the geometric construction can be represented by a ZXY Cardan chart. The labels ZXY alone do not specify clinical signs for both limbs.

The following listing deliberately computes the *mathematical chart* in Equation (2.3). It returns active right-hand angles in degrees, with the middle angle on  $(-90^\circ, 90^\circ)$  and outer angles on their principal atan2 branches. It does not automatically label those values as clinical flexion, adduction or internal rotation. Apply a documented anatomical calibration and side/sign mapping before making that interpretation. Input matrices must already be proper rotations; noisy marker data require a separately stated pose-estimation procedure.

```

1 import numpy as np
2 from numpy.typing import NDArray
3
4 def segment_zxy_angles(
5     R_proximal: NDArray[np.float64],
6     R_distal: NDArray[np.float64],
7 ) -> tuple[float, float, float]:
8     """Return chart angles; no clinical side/sign mapping is inferred."""
9     frames = []
10    for frame in (R_proximal, R_distal):
11        if np.iscomplexobj(frame):
12            raise ValueError("Expected real rotations")
13        R = np.asarray(frame, dtype=float)
14        if R.shape != (3, 3) or not np.all(np.isfinite(R)):
15            raise ValueError("Expected finite 3-by-3 rotations")
16        if not np.allclose(R.T @ R, np.eye(3), atol=1e-9, rtol=0):
17            raise ValueError("Rotation columns must be orthonormal")
18        if not np.isclose(np.linalg.det(R), 1, atol=1e-9, rtol=0):
19            raise ValueError("Rotation must be right-handed")
20        frames.append(R)
21    R = frames[0].T @ frames[1]
22    cos_beta = np.hypot(R[0, 1], R[1, 1])
23    if cos_beta < 1e-8:
24        raise ValueError("ZXY chart is singular or too ill-conditioned")
25    alpha = np.arctan2(-R[0, 1], R[1, 1])
26    beta = np.arctan2(R[2, 1], cos_beta)
27    gamma = np.arctan2(-R[2, 0], R[2, 2])
28    angles = np.degrees([alpha, beta, gamma])
29    return float(angles[0]), float(angles[1]), float(angles[2])

```

Listing 2.1: Extracting a declared active ZXY chart from segment orientations.

The singularity threshold is an illustrative numerical guard, not a sensor accuracy specification. Near the singular set, a small orientation error can produce a large angle error. Crossing an outer-angle branch cut can also create a 360° reporting jump without a comparable physical movement.

## 2.2.2 Multi-Representation Joint Angles

Following the multi-representation philosophy of this series, we express joint rotations and rigid displacements in complementary representations:

**1. Declared Active Intrinsic ZXY Angles:** Write  $R_{ij} = R_{S_i S_j}$  for the distal-to-proximal coordinate map:

$$R_{ij} = R_z(\alpha) R_x(\beta) R_y(\gamma), \quad (2.3)$$

Here the three angles are mathematical coordinates with right-hand positive rotations. They require the mapping discussed above before receiving clinical names. For (60°, 5°, 10°) the listing recovers the same triple. The middle sine is the entry in row 3, column 2, written  $R[2, 1]$  in Python; negating it would instead return -5°.

**2. Rotation Matrix (SO(3)):** Use  $[a]_{\times} b = a \times b$  for the skew map:

$$R_{ij} = \exp([\omega]_{\times} \theta) \in \text{SO}(3), \quad (2.4)$$

For a rotation,  $\omega$  is a unit axis and  $\theta$  an angle. A matrix stores nine entries with six independent orthonormality constraints and the additional choice  $\det R = +1$ . It avoids an

Euler chart's gimbal lock, but its axis-angle logarithm is not globally unique; the axis is undefined at identity.

### 3. Unit Quaternion:

$$\mathbf{q} = (\cos \frac{\theta}{2}, \boldsymbol{\omega} \sin \frac{\theta}{2}), \quad \|\mathbf{q}\| = 1, \quad (2.5)$$

A unit quaternion avoids Euler-chart gimbal lock, but  $\mathbf{q}$  and  $-\mathbf{q}$  represent the same rotation. Choose consistent signs for interpolation and state angle branches when converting back. Neither a quaternion nor a matrix removes singularities of the requested clinical-angle chart.

### 4. Screw Axis (Primary):

$$T_{ij} = e^{[\mathcal{S}]\theta} \in \text{SE}(3), \quad \mathcal{S} = (\boldsymbol{\omega}, \mathbf{v}) \in \mathbb{R}^6, \quad (2.6)$$

For a finite displacement with nonzero rotation, normalize  $\|\boldsymbol{\omega}\| = 1$ ; then  $\theta$  is a rotation angle and  $\mathbf{v}$  encodes the axis location and translation per unit angle. Here  $[\mathcal{S}]$  denotes the  $4 \times 4$  twist matrix with upper blocks  $[\boldsymbol{\omega}]_{\times}$  and  $\mathbf{v}$  and a zero bottom row. For pure translation use  $\boldsymbol{\omega} = 0$ ,  $\|\mathbf{v}\| = 1$  and a displacement parameter in units of length. Identity has no uniquely determined screw axis. Rotation alone does not determine  $T_{ij}$ : its translation must also be supplied.

#### Biology vs. Engineering 2.1. Why Screw Axis Theory Is Natural for Biological Joints

A biological articulation need not obey an ideal fixed revolute constraint. Relative translation and rotation can both matter, with their relationship dependent on joint geometry, loading and the modeled task. A finite screw describes the displacement between two configurations; an instantaneous screw axis describes a velocity. They need not coincide, and either can change during movement. At zero angular velocity an ordinary finite-pitch instantaneous axis is unavailable; pure translation is a distinct case.

The rigid-displacement framework in Lynch and Park's *Modern Robotics* [18] is useful for representing these motions. Representation alone supplies neither a biological joint law nor evidence that a fixed one-degree-of-freedom screw model is adequate.

## 2.3 Body Segment Parameters

### 2.3.1 Regression Equations

Body segment parameters (BSPs) include mass, center-of-mass location and the inertia tensor about a specified point and axes. A parameter table is tied to its segment boundaries, reference landmarks, measurement/modeling method and sample population. Examples illustrate why these conditions matter:

1. **de Leva (1996)** [6] adjusts the reference points of Zatsiorsky–Seluyanov parameters derived from young-adult male and female samples. This is a population-based adjustment, not direct measurement of every modeled subject or a universal gold standard.
2. **Pavol et al. (2002)** [24] describe an estimation method for older adults that incorporates body measurements and existing data sources. Its validation scope must accompany its use.

3. **Jensen (1986)** [15] studies growth-related changes in boys using a longitudinal sample and a geometric body model. Pediatric parameters should not be assumed to equal scaled adult values.

Distinguish a mean proportion from a subject-specific estimate and propagate plausible parameter uncertainty into the output of interest. Transferring a center-of-mass fraction between different segment endpoints changes its physical meaning. Likewise, a diagonal inertia tensor in principal axes need not remain diagonal in an anatomical frame. Changing coordinate axes rotates the tensor; changing its reference point additionally requires the parallel-axis theorem. These operations do not compensate for an inappropriate population or an inaccurate segment model.

### 2.3.2 Computing the Mass Matrix

Given segment inertial properties, the mass matrix  $M(\mathbf{q}) \in \mathbb{R}^{n \times n}$  of the multi-body system is computed using the composite rigid-body algorithm (see Volume 0's recursive-dynamics treatment). Use angular-first motion vectors  $V_i = (\boldsymbol{\omega}_i, \mathbf{v}_{O_i})$ , where  $\mathbf{v}_{O_i}$  is the segment-origin velocity expressed in that segment's frame. This is the body twist convention; a spatial twist's linear block in a fixed laboratory frame is generally not the segment-origin velocity. In the same body frame, the spatial inertia is [9]:

$$\mathcal{G}_i = \begin{bmatrix} I_i & m_i[\mathbf{c}_i]_{\times} \\ m_i[\mathbf{c}_i]_{\times}^T & m_i\mathbb{I}_3 \end{bmatrix} \in \mathbb{R}^{6 \times 6}, \quad (2.7)$$

where  $I_i$  is the  $3 \times 3$  rotational inertia about the segment frame origin,  $m_i$  is the segment mass,  $\mathbf{c}_i$  is the center of mass position in the segment frame directed from its origin to the COM, and  $[\mathbf{c}]_{\times} \mathbf{a} = \mathbf{c} \times \mathbf{a}$ . If the supplied rotational inertia is instead the central tensor  $I_{C_i}$ , first form

$$I_i = I_{C_i} + m_i(\|\mathbf{c}_i\|^2 \mathbb{I}_3 - \mathbf{c}_i \mathbf{c}_i^T). \quad (2.8)$$

Do not add that term again when  $I_i$  is already about the origin.

The off-diagonal signs follow from point transport,  $\mathbf{v}_{C_i} = \mathbf{v}_{O_i} + \boldsymbol{\omega}_i \times \mathbf{c}_i$ :

$$\frac{1}{2} V_i^T \mathcal{G}_i V_i = \frac{1}{2} \boldsymbol{\omega}_i^T I_{C_i} \boldsymbol{\omega}_i + \frac{1}{2} m_i \|\mathbf{v}_{O_i} + \boldsymbol{\omega}_i \times \mathbf{c}_i\|^2. \quad (2.9)$$

A symmetric positive-definite matrix alone is not sufficient verification: reversing both cross blocks is equivalent to using the opposite COM offset and can still give a positive-definite matrix. Check energy and momentum against point transport. Physical central principal moments are nonnegative and satisfy triangle inequalities; nondegenerate central inertia gives positive definiteness, while point/line idealizations can be semidefinite.

If  $V_i = J_i(\mathbf{q})\dot{\mathbf{q}}$  with consistent origins and axes, summing segment energies gives  $M(\mathbf{q}) = \sum_i J_i^T \mathcal{G}_i J_i$  for this rigid-segment model. Euler-angle rates are not Cartesian angular-velocity components. Generalized forces follow from power: a segment wrench  $F_i$  contributes  $J_i^T F_i$ , not an arbitrary selection of its Cartesian entries. For a golf swing, for example, an apparent change in joint angles or inferred moments can reflect calibration, parameter or coordinate choices as well as a change in the swing. A coordinate correction is not evidence of a different muscle or neural strategy.

## Exercises

1. **Coordinate System Verification.** Verify that the declared ZXY listing returns zero for aligned frames and recovers mixed positive and negative chart angles. Premultiply both input frames by the same rotation and verify invariance. Specify the additional calibration and side/sign mapping needed for clinical labels.
2. **Euler Angle Singularity.** Identify the singular ZXY configurations and explain why they depend on the middle angle, not on flexion magnitude alone. Perturb a near-singular rotation and quantify angle sensitivity. Distinguish this chart problem from a physiological restriction; no claim about gait or squat ranges follows from the algebra alone.
3. **Multi-Representation Conversion.** Write Python functions to convert between all four representations (Euler ZXY, rotation matrix, quaternion, screw displacement) for chart angles ( $60^\circ, 5^\circ, 10^\circ$ ) and a separately specified translation  $\mathbf{p} = (0.01, 0, 0)^T$  m. State angle/logarithm branches and check reconstruction. Explain why the three angles alone are insufficient for the rigid-displacement conversion.
4. **BSP Sensitivity.** For an isolated horizontal rigid segment of mass 4 kg, compare COM distances 0.20 m and 0.22 m from a fixed support using  $g = 9.81 \text{ m/s}^2$ . Compute the balancing moment magnitudes and percentage change. These are manufactured parameter choices, not two published population estimates. Explain why the net support moment does not identify muscle force without moment arms and load-sharing assumptions.
5. **(Programming.)** Implement the spatial inertia matrix computation for a 3-segment leg model (thigh, shank, foot), using the companion repository's `affine_control.dynamics` function `spatial_inertia(mass, com, inertia_com)`. State its central-inertia input convention. Verify symmetry, physical central moments and kinetic energy under simultaneous rotation and translation, including a nonzero COM offset.
6. **(Programming.)** Using motion capture data (or simulated data), compute chart angles from segment orientations with the provided listing. Document calibration, units, invalid-frame treatment and branch handling. Plot a clinical flexion trace only after declaring and checking its anatomical mapping; identify simulated data as simulated.

## Chapter 3

# Muscle Models

A muscle is not a motor. It is a spring with a memory, a damper with an agenda, and a force generator with a personality.

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### 3.1 The Hill-Type Muscle Model

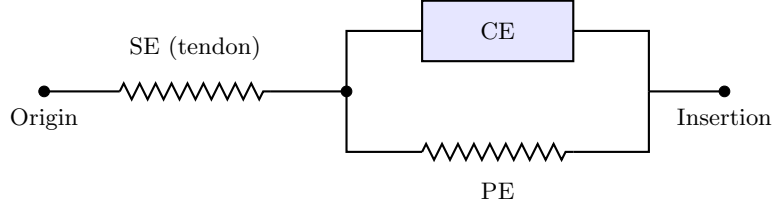
A joint moment is not yet a muscle model. To predict how excitation becomes force, we must specify activation, fiber length and velocity, tendon stretch, and the geometry through which tension acts on the skeleton. These quantities evolve together: joint motion changes the muscle path, while muscle tension changes joint motion.

Hill's shortening experiments motivate a force-velocity relation [14]; modern Hill-type models combine phenomenological constitutive curves with selected mechanical and physiological states. They form a family, not one universal equation. We use a massless, undamped equilibrium architecture, constant-thickness pennation, and explicitly declared fiber curves. Thelen's formulation and Millard's equilibrium, damped-equilibrium and rigid-tendon alternatives make different numerical choices [27, 19]. A force evaluator is not by itself a simulation of any of these complete systems.

#### 3.1.1 Model Architecture

The three-element network used here has the following components. The diagram shows mechanical connectivity, not literal fiber orientation:

1. **Contractile Element (CE)**: represents the active force-generating capacity of the muscle fibers. Its force depends on activation  $a(t)$ , fiber length  $\ell^M$ , and fiber velocity  $v^M = \dot{\ell}^M$ .
2. **Parallel Elastic Element (PE)**: represents the passive elastic properties of the muscle belly (titin filaments, connective tissue). In this particular curve, passive tension begins above optimal fiber length; that threshold is a modeling assumption, not a universal resting length.
3. **Series Elastic Element (SE)**: represents the tendon, which is in series with the parallel CE-PE combination. Modeled here as a lossless nonlinear spring with a slack length  $\ell_T^s$ .



### 3.1.2 The Force-Length Relationship

The isometric (zero velocity) force of the contractile element depends on the normalized fiber length  $\tilde{\ell} = \ell^M / \ell_0^M$ , where  $\ell_0^M$  is optimal fiber length. Let  $F_0$  be the full-activation, optimal-length isometric *active fiber* force scale:

#### Definition 3.1. Active Force-Length Curve

The normalized active force-length relationship is:

$$f_L(\tilde{\ell}) = \exp\left(-\frac{(\tilde{\ell} - 1)^2}{\gamma}\right), \quad (3.1)$$

where  $\gamma \approx 0.45$  [27] controls the width of the curve. Maximum force ( $f_L = 1$ ) occurs at optimal fiber length ( $\tilde{\ell} = 1$ ). This Gaussian is strictly positive at finite lengths. At both  $\tilde{\ell} = 0.5$  and  $1.5$  it gives  $0.574$ , not nearly zero. The half-height interval is  $|\tilde{\ell} - 1| < \sqrt{\gamma \ln 2}$ , approximately  $(0.442, 1.558)$  for this width. These are properties of the approximation, not measured anatomical operating limits.

*Parameter and evidence scope.* Thelen's Appendix uses  $\gamma = 0.45$ ,  $A_f = 0.25$ ,  $k^{PE} = 5$ , passive reference strain  $\varepsilon_0 = 0.6$ , and the activation-dependent law in (3.3) [27]. These are model parameters, not universal human constants. Millard uses separately constructed smooth curves rather than establishing these numbers as shared defaults [19].

Filament overlap helps explain length-dependent active tension. Gordon, Huxley and Julian studied tetanized single frog fibers and found a plateau and distinct ascending and descending regions [10]. That evidence supports length dependence; it does not identify this Gaussian as a microscopic law or validate its extrapolation to every human muscle.

The passive curve is

$$f_{PE}(\tilde{\ell}) = \begin{cases} 0, & 0 < \tilde{\ell} \leq 1, \\ \frac{\exp[k^{PE}(\tilde{\ell} - 1)/\varepsilon_0] - 1}{\exp(k^{PE}) - 1}, & \tilde{\ell} > 1. \end{cases} \quad (3.2)$$

It is continuous, nonnegative, and equals one at  $\tilde{\ell} = 1 + \varepsilon_0$ . Its derivative changes at the slack threshold. The exponential has no finite global maximum, so a force surface needs a declared length domain.

### 3.1.3 A Constitutive Evaluator With Declared Inputs

The listing takes fiber length and velocity as inputs and returns tendon-axis projections. It does *not* infer those inputs from joint motion or enforce tendon equilibrium. The stored tendon slack length becomes relevant when the series element and path geometry are supplied. The biceps force and length scales reproduce BIClong in the pinned OpenSim Arm26 model [22]; they do not describe both biceps heads or an individual golfer. The full-activation speed scale is explicitly chosen as  $10\ell_0^M/s$  for this example.

```

1 from dataclasses import dataclass
2 import math
3
4 def finite_values(values: tuple[float, ...]) -> None:
5     """Reject nonfinite scalar inputs."""
6     if not all(math.isfinite(value) for value in values):
7         raise ValueError("Inputs must be finite.")
8
9 @dataclass(frozen=True)
10 class MuscleParameters:
11     """Fiber constitutive parameters; SI units, radians."""
12     F0: float
13     l0_M: float
14     v0: float
15     l_T_slack: float
16     alpha0: float
17     gamma: float = 0.45
18     Af: float = 0.25
19     kPE: float = 5.0
20     e0: float = 0.6
21
22     def __post_init__(self) -> None:
23         values = (self.F0, self.l0_M, self.v0,
24                 self.l_T_slack, self.gamma,
25                 self.Af, self.kPE, self.e0)
26         finite_values((*values, self.alpha0))
27         if min(values) <= 0 or not 0 <= self.alpha0 < math.pi / 2:
28             raise ValueError("Positive scales and 0 <= alpha0 < pi/2 required
29                               .")
30
31     @property
32     def v_max(self) -> float:
33         """Full-activation speed scale in fiber lengths/s."""
34         return self.v0 / self.l0_M
35
36 def force_length_active(l_tilde: float, gamma: float = 0.45) -> float:
37     """Normalized active fiber force at zero velocity."""
38     finite_values((l_tilde, gamma))
39     if l_tilde <= 0 or gamma <= 0:
40         raise ValueError("Positive length and width required.")
41     return math.exp(-((l_tilde - 1.0) ** 2) / gamma)
42
43 def force_length_passive(l_tilde: float, kPE: float = 5.0,
44                         e0: float = 0.6) -> float:
45     """Passive fiber tension, normalized by F0."""
46     finite_values((l_tilde, kPE, e0))
47     if l_tilde <= 0 or kPE <= 0 or e0 <= 0:
48         raise ValueError("Positive length and shape scales required.")
49     if l_tilde <= 1.0:
50         return 0.0
51     try:

```

```

51         return math.exp(1/(kPE * (l_tilde - 1.0) / e0)) / math.exp(1/(kPE *
52     except OverflowError as error:
53         raise ValueError("Constitutive extrapolation exceeds numeric range.")
54     from error
55
56 def force_velocity(v_tilde: float, a: float, Af: float = 0.25,
57     Flen: float = 1.4) -> float:
58     """Thelen interior multiplier; v_tilde = v_M / v0."""
59     finite_values((v_tilde, a, Af, Flen))
60     if not 0 <= a <= 1 or Af <= 0 or Flen <= 1:
61         raise ValueError("Require a in [0,1], Af > 0, Flen > 1.")
62     speed = 0.25 + 0.75 * a
63     if v_tilde < -speed:
64         raise ValueError("Shortening exceeds the declared interior branch.")
65     w = v_tilde / speed
66     if w <= 0:
67         return (1.0 + w) / (1.0 - w / Af)
68     shape = (2.0 + 2.0 / Af) / (Flen - 1.0)
69     # Algebraic rearrangement avoids multiplying two large velocities.
70     return Flen - (Flen - 1.0) / (1.0 + shape * w)
71
72 def muscle_force(activation: float, l_M: float, v_M: float,
73     params: MuscleParameters) -> tuple[float, float, float]:
74     """Tendon-axis projections for PRESCRIBED fiber state, in N.
75
76     Returns total, active and passive projections. This does not
77     solve tendon equilibrium or determine the fiber state.
78     """
79     finite_values((activation, l_M, v_M))
80     if not 0 <= activation <= 1:
81         raise ValueError("Activation must lie in [0,1].")
82     thickness = params.l0_M * math.sin(params.alpha0)
83     if l_M <= thickness:
84         raise ValueError("Require fiber length greater than thickness.")
85     cos_alpha = math.sqrt(1.0 - (thickness / l_M) ** 2)
86     length = l_M / params.l0_M
87     active = (activation * force_length_active(length, params.gamma)
88         * force_velocity(v_M / params.v0, activation, params.Af))
89     passive = force_length_passive(length, params.kPE, params.e0)
90     scale = params.F0 * cos_alpha
91     result = ((active + passive) * scale, active * scale, passive * scale)
92     finite_values(result)
93     return result
94
95 # Long-head biceps geometry/force scales from OpenSim Arm26 BICLong.
96 # v0 and curve parameters are declared teaching choices.
97 biceps = MuscleParameters(F0=624.3, l0_M=0.1157,
98     v0=10 * 0.1157,
99     l_T_slack=0.2723, alpha0=0.0)
100 total, active, passive = muscle_force(0.5, biceps.l0_M, 0.0, biceps)
101 print(f"Prescribed isometric fiber: {total:.2f} N "
102     f"(active {active:.2f}, passive {passive:.2f})")

```

Listing 3.1: Prescribed-Fiber Constitutive Evaluation.

### 3.1.4 The Force-Velocity Relationship

Let  $v_0 > 0$  be the dimensional full-activation shortening-speed scale and  $\tilde{v} = v^M/v_0$ , with negative velocity denoting shortening. Here  $A_f = 0.25$  controls the shortening branch and

$F_{\text{len}} = 1.4$  is the dimensionless lengthening asymptote used by the listing. For the interior branches obtained from Thelen's equations, define

$$c(a) = 0.25 + 0.75a, \quad w = \tilde{v}/c(a), \quad K = \frac{2 + 2/A_f}{F_{\text{len}} - 1}.$$

Solving the velocity-as-a-function-of-force relation for its multiplier gives

$$f_V(\tilde{v}, a) = \begin{cases} \frac{1 + w}{1 - w/A_f}, & -1 \leq w \leq 0, \\ \frac{1 + KF_{\text{len}}w}{1 + Kw}, & w > 0. \end{cases} \quad (3.3)$$

Thus  $f_V(0, a) = 1$ , unloaded shortening occurs at  $v^M = -c(a)v_0$ , and the lengthening branch approaches  $F_{\text{len}}$  asymptotically. At half activation the zero-force speed is  $-0.625v_0$ , not  $-v_0$ . The multiplier is continuous at zero velocity, but the right slope is twice the left slope. An optimizer must not assume a continuously differentiable join.

This evaluator rejects shortening outside its declared branch rather than returning negative active tension. The current OpenSim implementation adds continuations and state restrictions [23]; this listing is not a replacement for that solver. An inverse equilibrium calculation also requires  $af_L > 0$  and an attainable force ratio below  $F_{\text{len}}$ . At zero activation the forward active force is zero, but dividing by activation to recover velocity is undefined.

#### Biomechanical Principle 3.1. Force, Velocity, and Power

Shortening reduces active force below its isometric value; lengthening increases the multiplier above one, at the same activation and fiber length. This does not say that eccentric force always exceeds the full-activation, optimal-length reference force  $F_0$ . Isometric *fiber* force includes active and passive contributions. Zero fiber velocity need not mean zero joint velocity; fixed joints need not imply isometric fibers while a tendon stretches. Active mechanical output is  $-F_{CE}v^M$ , with  $F_{CE}$  the active fiber force,, positive during shortening and negative during lengthening. Its sign does not determine metabolic expenditure.

### 3.1.5 Activation Dynamics

The neural command from the central nervous system does not instantly produce muscle force. The activation dynamics models the excitation-contraction coupling:

$$\dot{a}(t) = \frac{u(t) - a(t)}{\tau(u, a)}, \quad (3.4)$$

where  $u(t) \in [0, 1]$  is the neural excitation,  $a(t) \in [0, 1]$  is the activation level, and  $\tau$  is a time constant that differs for activation and deactivation:

$$\tau(u, a) = \begin{cases} \tau_{\text{act}}(0.5 + 1.5a) & \text{if } u > a \text{ (activation),} \\ \tau_{\text{deact}}/(0.5 + 1.5a) & \text{if } u \leq a \text{ (deactivation).} \end{cases} \quad (3.5)$$

The exercise uses  $\tau_{\text{act}} = 15$  ms and  $\tau_{\text{deact}} = 50$  ms, as in the young-adult parameter set of [27]. These base parameters are not observed force-rise or neural-conduction delays. At

$a = 1$  the two effective branch times are 30 and 25 ms, respectively; the base constants do not imply faster activation at every state.

The vector field is continuous at  $u = a$  because both branches give zero there; its derivatives generally differ across that surface. Continuous-time trajectories preserve  $[0, 1]$ : the derivative points inward at its boundaries. Numerical integration still needs suitable tolerances and checked states.

With  $a(0) = 0$  and a step to  $u = 1$ , separation of variables gives

$$t(a) = \tau_{\text{act}} \{-1.5a - 2 \ln(1 - a)\}.$$

Activation reaches 0.95 after approximately 68.50 ms. For deactivation from  $a(0) = 1$  with  $u = 0$ ,

$$t(a) = 2\tau_{\text{deact}} \ln\left(\frac{0.5 + 1.5a}{2a}\right),$$

so  $a = 0.05$  occurs at approximately 174.92 ms. These are activation-state calculations; tendon force follows the coupled fiber and series-elastic dynamics.

### 3.2 Tendon Mechanics

The tendon connects the muscle belly to bone. It is modeled as a nonlinear elastic element with a resting (slack) length  $\ell_T^s$ :

$$F_T(\varepsilon_T) = \begin{cases} 0 & \varepsilon_T \leq 0, \\ F_0 \cdot \frac{e^{k_T \varepsilon_T} - 1}{e^{k_T \varepsilon_{0T}} - 1} & \varepsilon_T > 0, \end{cases} \quad (3.6)$$

where  $\ell_T$  is current tendon length and  $\varepsilon_T = (\ell_T - \ell_T^s)/\ell_T^s$ ,  $k_T > 0$  and  $\varepsilon_{0T} > 0$ . This single-exponential tendon is a *declared teaching choice*, not Thelen's toe-region/linear tendon or a full Millard curve. It gives  $F_T = F_0$  at the reference strain for *every*  $k_T$ ; that reference point cannot identify the shape parameter. A second measurement, such as tangent stiffness, is needed:

$$\frac{dF_T}{d\ell_T} = \frac{F_0 k_T e^{k_T \varepsilon_T}}{\ell_T^s (e^{k_T \varepsilon_{0T}} - 1)} \quad (\varepsilon_T > 0).$$

For  $F_T > 0$ , its inverse is

$$\varepsilon_T = \frac{1}{k_T} \ln \left[ 1 + \frac{F_T}{F_0} (e^{k_T \varepsilon_{0T}} - 1) \right].$$

Force is continuous at slack engagement but stiffness changes there.

#### Biomechanical Principle 3.2. Tendons as Energy Storage Elements

The lossless teaching spring stores  $U_T = \ell_T^s \int_0^{\varepsilon_T} F_T(e) de$  above slack. It returns that energy on unloading; it creates none. Biological viscoelasticity can dissipate energy and needs an additional constitutive law.

Elastic recoil can separate the timing of fiber work from joint power. Alexander's account concerns locomotor mechanisms [2]; it does not establish a metabolic saving or tendon-recoil strategy in a particular golf swing. Larger instantaneous joint power

alone cannot identify its muscular, elastic, or external source.

### 3.3 Musculotendon Equilibrium

Neglecting fiber and series-element inertia, the declared axial balance is

$$F_T = (F_{CE} + F_{PE}) \cos \alpha, \quad F_{CE} = a f_L f_V F_0, \quad F_{PE} = f_{PE} F_0. \quad (3.7)$$

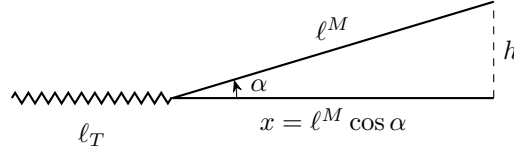
This is a model assumption, not a claim that real accelerating muscle has no inertia. Any added damping force belongs in this balance and the energy account.

Let  $\alpha$  be current pennation angle and  $\alpha_0$  its value at optimal fiber length. For constant fiber-bundle thickness  $h = \ell_0^M \sin \alpha_0$  and total musculotendon path length  $\ell^{MT}$ ,

$$x = \ell^M \cos \alpha = \sqrt{(\ell^M)^2 - h^2}, \quad \ell^{MT} = \ell_T + x, \quad \ell^M > h. \quad (3.8)$$

An impossible length is not repaired by clipping  $\cos \alpha$  to zero. The limit  $\ell^M = h$  is singular.

Constant-Thickness Geometry



Given activation, fiber length and total path length, compute tendon length and tension, then solve for fiber velocity. Define the required active-force multiplier

$$r = \frac{F_T / (F_0 \cos \alpha) - f_{PE}}{a f_L}.$$

The independent inverse of the declared interior law is

$$\frac{v^M}{v_0} = \begin{cases} c(a) \frac{r-1}{1+r/A_f}, & 0 \leq r \leq 1, \\ c(a) \frac{(r-1)(F_{\text{len}}-1)}{(2+2/A_f)(F_{\text{len}}-r)}, & 1 < r < F_{\text{len}}. \end{cases} \quad (3.9)$$

Negative residual active-force demand, a ratio at or above the eccentric asymptote, zero activation, or zero active-length multiplier lies outside this inverse. Near the asymptote small force errors imply large velocity changes. A solver must report failure or use an explicitly different regularized model, not silently clamp the answer.

#### 3.3.1 Initialization and Integration

Fiber length is a state. Activation and total path length alone do not determine both fiber length and velocity. Specify  $a(0)$  and  $\ell^M(0)$ , or an explicit initialization rule. For an assumed initially isometric fiber, search for a root of (3.7) with  $v^M = 0$  in an admissible interval.

The nonlinear curves may admit multiple roots or none; record the interval, residual and selected root rather than assuming uniqueness.

At each evaluation:

1. Check states and geometry; derive  $\ell_T = \ell^{MT} - x$ .
2. Evaluate tendon and passive-fiber forces.
3. Check  $af_L \cos \alpha$  and the attainable ratio, then invert for  $v^M$ .
4. Integrate activation and fiber length with error control, monitoring the force residual and domain boundaries.

For this undamped inverse, a slack tendon with positive passive fiber tension has no admissible nonnegative active-force balance. Damped or constrained models can behave differently; their equations and state policy must be declared.

### 3.4 From Fiber Power to Joint Mechanics

With  $v^{MT} = \dot{\ell}^{MT}$  and  $v_T = \dot{\ell}_T$ , differentiating the geometry gives

$$v^{MT} = v_T + \frac{v^M}{\cos \alpha}.$$

The force projection then gives  $F_T(v^{MT} - v_T) = (F_{CE} + F_{PE})v^M$ . Projecting force and also multiplying fiber velocity by  $\cos \alpha$  would count the geometry incorrectly.

Let  $\mathbf{q}$  be generalized coordinates and  $\mathbf{Q}_M$  the generalized muscle load. For a frictionless path  $\ell^{MT}(\mathbf{q})$ , virtual work gives

$$\mathbf{Q}_M = -F_T \nabla_{\mathbf{q}} \ell^{MT}, \quad \mathbf{Q}_M^T \dot{\mathbf{q}} = -F_T v^{MT}.$$

For lossless tendon and passive fiber, let  $U_{PE}$  be passive-fiber stored energy, so  $\dot{U}_{PE} = F_{PE}v^M$ . Then

$$\mathbf{Q}_M^T \dot{\mathbf{q}} = -F_{CE}v^M - \dot{U}_{PE} - \dot{U}_T. \quad (3.10)$$

Active output and changing stored energy jointly produce skeletal power. A tendon can deliver power with nearly isometric fibers, or absorb power while active fibers shorten. Neither observation uniquely identifies excitation, chemical-energy use, or the cause of segment deceleration. Frictional wraps would require an additional dissipation account.

#### 3.4.1 Coactivation, Equilibrium, and Stiffness

Index muscles by  $i$ , with tendon force  $F_i$  and path length  $\ell_i^{MT}$ . For a joint coordinate  $q$ , define  $\ell'_i = d\ell_i^{MT}/dq$  and  $\ell''_i = d^2\ell_i^{MT}/dq^2$ . Write joint moment as  $\tau_q$  to distinguish it from the activation time scale. A declared quasistatic path-force law at fixed activation gives

$$\tau_q = -\sum_i F_i \ell'_i, \quad K_q = -\frac{d\tau_q}{dq} = \sum_i [k_i (\ell'_i)^2 + F_i \ell''_i], \quad k_i = \frac{dF_i}{d\ell_i^{MT}}.$$

Material response and changing moment arms both contribute. Define how fiber and tendon equilibrate when calculating  $k_i$ . Frozen states, quasistatic relaxation and dynamic

impedance are different experiments; activation dynamics, frequency and feedback matter to the last.

Equal excitation of unequal antagonists does not generally balance their moments. Symmetric force capacities and opposite matched moment arms can preserve equilibrium under balanced coactivation, but that is a controlled example, not a conclusion about biceps and triceps. The active descending length branch and geometric term also prevent a universal positive-stiffness guarantee.

### 3.5 The Hill Model as a Coupled Dynamical System

For the declared equilibrium model,

$$\dot{a} = (u - a)/\tau(u, a), \quad (3.11)$$

$$\dot{\ell}^M = v^M(a, \ell^M, \ell^{MT}), \quad (3.12)$$

using (3.9) only on its admissible domain. Pennation, tendon geometry, constitutive ranges and invertibility restrict the state space; it is not the unrestricted product  $[0, 1] \times \mathbb{R}_{>0}$ .

With prescribed joint motion, path length is an input signal. In forward dynamics it is  $\ell^{MT}(\mathbf{q})$  inside a feedback coupling: excitation changes muscle state, tension changes skeletal loads, and motion changes muscle length. Zero net joint moment therefore does not imply inactive muscles, and identical joint trajectories can coexist with different coactivation and stored-energy histories.

#### Biomechanical Principle 3.3. Model Choice Follows the Question

Distinguish prescribed-fiber evaluation, compliant equilibrium simulation, damped equilibrium simulation, and rigid-tendon approximation. Each answers a different mechanical question.

Millard and colleagues compared computational behavior and force fits in particular preparations, including rat and cat soleus [19]. Those results do not rank every model for every human task. Validation should name the muscle, population, loading protocol, predicted quantity, parameter estimation and independent test data. A fit to swing kinematics cannot alone validate individual muscle force.

### Exercises

1. **Force-Length Range.** Plot  $f_L$  for  $\gamma \in \{0.3, 0.45, 0.6\}$  and derive the half-height interval. Explain why its endpoints are not anatomical limits. Independently check passive continuity, nonnegativity and  $f_{PE}(1 + \varepsilon_0) = 1$ .
2. **Velocity and Its Inverse.** Plot both branches at  $a = 0.5$  and 1. Mark unloaded shortening, the isometric point and the unattained lengthening asymptote. Verify forward/inverse agreement at  $r = 0.4$  and 1.2. Compare one-sided slopes at zero velocity.
3. **Activation Timing.** Use the stated 15/50 ms parameters,  $a(0) = 0$ , and  $u = 1$  for 200 ms followed by  $u = 0$ . Compare integration with the analytic solutions. Measure

decay relative to the actual state at switch-off, distinguishing it from decay starting at one.

4. **Tendon Shape and Energy.** Use a synthetic teaching tendon with  $\ell_T^s = 0.25$  m,  $F_0 = 4000$  N,  $\varepsilon_{0T} = 0.04$  and  $k_T = 35$ . Compute stretch at  $0.5F_0$ ,  $0.75F_0$  and  $F_0$ , and integrate stored energy. Repeat with  $k_T = 10$ . Explain why both laws share the reference force but have different stiffness. These choices do not calibrate an Achilles tendon.
5. **Force Surface.** Plot active, passive and total *fiber* force over  $0.5 \leq \tilde{\ell} \leq 1.6$  and  $0 \leq a \leq 1$ , at zero fiber velocity and zero pennation. Locate bounded-domain maxima. Explain why extrapolating the passive exponential to unbounded length gives no finite global total-force maximum.
6. **Equilibrium Simulation.** Implement the checked procedure for a prescribed path and excitation. Declare initial states, constitutive laws, root selection and domain policy. Verify force residual and power balance; report step-size convergence. Demonstrate a clear failure for an inadmissible state rather than a clipped trajectory.
7. **Antagonists and Geometry.** Begin with matched antagonists, opposite constant moment arms and a balanced equilibrium. Calculate torque and tangent stiffness as common activation changes. Then change one force capacity and introduce curved paths. Identify the symmetry assumptions needed to preserve equilibrium; distinguish tangent stiffness from impedance.
8. **Model Comparison.** Compare specified Thelen and Millard implementations using the same data and initialization. Report force error, solver cost and domain failures. Distinguish numerical regularization from physiological validation and identify evidence needed for a golf-swing claim.

## Chapter 4

# Joint Kinematics and Soft Tissue

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**Draft Notice.** This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

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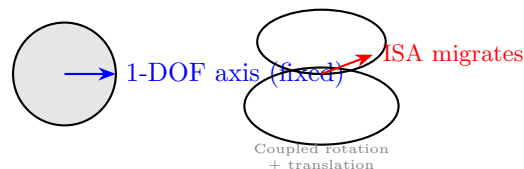
The knee is not a hinge. The shoulder is not a ball-and-socket. These are useful fictions that biology tolerates but does not enforce.

---

### 4.1 Beyond Ideal Joints

Engineering joints are designed to constrain motion to specific degrees of freedom: a revolute joint permits one rotation, a prismatic joint permits one translation. Biological joints achieve constraint through a fundamentally different mechanism: the interaction of articular surfaces, ligaments, joint capsule, and muscle forces. This section develops the kinematics of biological joints using the screw theory framework of Volume 0, extended to handle the additional complexity.

#### Engineering Revolute Biological Joint



#### 4.1.1 The Instantaneous Screw Axis

Every rigid-body motion can be decomposed into a rotation about and translation along a **screw axis** (Chasles' theorem, Volume 0, Chapter 5). For an engineering revolute joint, this screw axis is fixed in both bodies. For a biological joint, the instantaneous screw axis (ISA) *moves* as the joint configuration changes.

**Definition 4.1. Instantaneous Screw Axis of a Biological Joint**

Given the relative twist  $\mathcal{V}_{ij} = (\boldsymbol{\omega}, \mathbf{v}) \in \mathbb{R}^6$  between adjacent segments  $i$  and  $j$ , the instantaneous screw axis has:

- Direction:  $\hat{\mathbf{s}} = \boldsymbol{\omega} / \|\boldsymbol{\omega}\|$
- Pitch:  $h = \boldsymbol{\omega} \cdot \mathbf{v} / \|\boldsymbol{\omega}\|^2$
- Location:  $\mathbf{q} = \boldsymbol{\omega} \times \mathbf{v} / \|\boldsymbol{\omega}\|^2$

For a pure rotation ( $h = 0$ ), the ISA passes through  $\mathbf{q}$  in direction  $\hat{\mathbf{s}}$ . Tracking how  $\mathbf{q}$  and  $\hat{\mathbf{s}}$  change with joint angle reveals the coupled kinematics of the biological joint.

**4.1.2 The Knee: Rolling, Sliding, and Coupled Rotation**

The tibiofemoral joint exemplifies the complexity of biological kinematics. During flexion from  $0^\circ$  to  $90^\circ$ :

1. The femoral condyles **roll** on the tibial plateaus (like a wheel on a road)
2. They simultaneously **slide** posteriorly (the contact point moves backward)
3. The tibia undergoes coupled **internal rotation** of approximately  $15^\circ$  (the “screw-home” mechanism)
4. The contact area shifts, changing the effective moment arm

This coupled motion means the knee ISA migrates superiorly and posteriorly during flexion. The position and velocity of the ISA are key outputs of kinematic analysis [5, 30]. The four-bar linkage model provides a simplified but insightful mechanical analog for understanding coupled knee kinematics [21]:

```

1 import numpy as np
2 from typing import Tuple
3
4 def compute_isa(
5     omega: np.ndarray,
6     v: np.ndarray
7 ) -> Tuple[np.ndarray, np.ndarray, float]:
8     """Compute the instantaneous screw axis from a twist.
9
10    Parameters
11    -----
12    omega : np.ndarray, shape (3,)
13           Angular velocity vector.
14    v : np.ndarray, shape (3,)
15       Linear velocity of the body-fixed frame origin.
16
17    Returns
18    -----
19    Tuple[np.ndarray, np.ndarray, float]
20        (point_on_axis, axis_direction, pitch)
21    """
22    omega_norm = np.linalg.norm(omega)
23    if omega_norm < 1e-10:

```

```

24     # Pure translation
25     direction = v / np.linalg.norm(v)
26     return np.zeros(3), direction, np.inf
27
28     direction = omega / omega_norm
29     pitch = np.dot(omega, v) / omega_norm**2
30     point = np.cross(omega, v) / omega_norm**2
31
32     return point, direction, pitch
33
34
35 def knee_isa_trajectory(
36     flexion_angles: np.ndarray,
37     roll_slide_ratio: float = 0.6
38 ) -> np.ndarray:
39     """Estimate knee ISA position during flexion.
40
41     Simple model: ISA migrates posteriorly and superiorly
42     with flexion, governed by the roll-to-slide ratio.
43
44     Parameters
45     -----
46     flexion_angles : np.ndarray
47         Array of flexion angles in degrees.
48     roll_slide_ratio : float
49         Ratio of rolling to total contact displacement.
50         Pure rolling = 1.0, pure sliding = 0.0.
51
52     Returns
53     -----
54     np.ndarray, shape (N, 3)
55         ISA positions in the tibial reference frame.
56     """
57     R_condyle = 0.030 # approximate femoral condyle radius (m)
58     isa_positions = np.zeros((len(flexion_angles), 3))
59
60     for i, theta in enumerate(np.radians(flexion_angles)):
61         # Posterior migration due to rolling
62         posterior = R_condyle * theta * roll_slide_ratio
63         # Superior migration (ISA moves up with flexion)
64         superior = R_condyle * (1.0 - np.cos(theta * 0.5))
65
66         isa_positions[i] = [-posterior, superior, 0.0]
67
68     return isa_positions

```

Listing 4.1: Instantaneous screw axis computation for a biological joint.

### 4.1.3 The Shoulder: The Most Complex Joint

The glenohumeral joint has the largest range of motion of any joint in the body, achieved at the expense of stability. The “ball-and-socket” description is a gross simplification: quantitative measurements show the humeral head translates during abduction [16, 8], and the instantaneous center of rotation shifts continuously. The instantaneous screw axis of the shoulder is not fixed and migrates through the joint during arm motion [12].

The shoulder complex involves four joints:

1. **Glenohumeral (GH)**: primary ball-and-socket, 3 DOF rotation +  $\sim 3$  mm translation

2. **Acromioclavicular** (AC): 3 DOF rotation, small range
3. **Sternoclavicular** (SC): 3 DOF rotation, small range
4. **Scapulothoracic** (ST): not a true synovial joint; the scapula glides on the thorax, constrained as a *surface contact* rather than a pin joint

The scapulothoracic “joint” is particularly interesting from a mathematical perspective: it is a *holonomic constraint* that the scapula must remain on the thoracic surface, effectively acting as a 2-DOF constraint (the scapula can translate on the surface but cannot lift off).

## 4.2 Ligament Models

Ligaments constrain joint motion by resisting excessive displacement. They are modeled as nonlinear springs with:

$$F_{\text{lig}}(\varepsilon) = \begin{cases} 0 & \varepsilon \leq 0 \text{ (slack),} \\ k_1 \varepsilon^2 / (4\varepsilon_1) & 0 < \varepsilon \leq 2\varepsilon_1 \text{ (toe region),} \\ k_2(\varepsilon - \varepsilon_1) & \varepsilon > 2\varepsilon_1 \text{ (linear region),} \end{cases} \quad (4.1)$$

where  $\varepsilon = (\ell - \ell_0)/\ell_0$  is the strain,  $\ell_0$  is the reference length,  $\varepsilon_1 \approx 0.03$  is the transition strain, and  $k_1, k_2$  are stiffness parameters [3].

## 4.3 Spinal Mechanics

The spine presents a unique modeling challenge: 24 vertebrae connected by intervertebral discs, each joint having 6 coupled degrees of freedom. The functional spinal unit (FSU) consisting of two adjacent vertebrae and their connecting disc is the basic mechanical unit.

The intervertebral disc consists of:

- **Nucleus pulposus**: a gel-like center that distributes compressive loads
- **Annulus fibrosus**: concentric rings of collagen fibers that resist torsion and bending

A common simplified model treats each FSU as a 6-DOF spring:

$$\mathbf{F}_{\text{disc}} = K_{\text{disc}} \cdot \boldsymbol{\delta}, \quad (4.2)$$

where  $K_{\text{disc}} \in \mathbb{R}^{6 \times 6}$  is the stiffness matrix and  $\boldsymbol{\delta} \in \mathbb{R}^6$  is the generalized displacement (3 translations + 3 rotations).

## Exercises

1. **ISA Computation.** For a knee flexing at  $60^\circ/\text{s}$  with 2 mm/s of posterior translation, compute the ISA direction, pitch, and location.
2. **Screw-Home Mechanism.** Plot the coupled internal rotation of the tibia as a function of knee flexion using the model:  $\gamma = 15^\circ \cdot (1 - \cos(\theta/2))$  for  $\theta \in [0, 90^\circ]$ .

3. **Shoulder Kinematics.** Using screw axis theory, express the configuration of the hand (end-effector) as a product of exponentials through the SC, AC, GH, and elbow joints.
4. **(Programming.)** Implement the ligament force model and plot the force-strain curve. Identify the toe region and the linear region.
5. **(Programming.)** Create a 3D visualization of the knee ISA trajectory during flexion from  $0^\circ$  to  $120^\circ$ .

## Chapter 5

# Multi-Body Dynamics of Biological Chains

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**Draft Notice.** This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

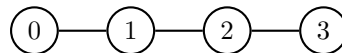
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The human body is a high-dimensional articulated system with changing contacts and compliant tissues. Rigid-body dynamics is a useful approximation, but never the whole story.

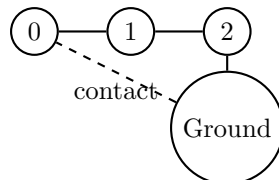
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### 5.1 Open-Chain vs. Closed-Chain Biomechanics

Most human movement involves open kinematic chains (the limbs) connected to a floating base (the pelvis/trunk). However, many functional activities create temporary closed chains: the stance phase of gait (foot on ground), pushing against a wall, or the golf swing when both hands grip the club.



**Open Chain** (arm reaching)



**Closed Chain** (gait stance phase)

### 5.1.1 Equations of Motion for Biological Chains

The equations of motion for an  $n$ -DOF musculoskeletal system take the standard Lagrangian form:

$$M(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + g(\mathbf{q}) = R(\mathbf{q})\mathbf{F}_{\text{muscle}} + J_c^T(\mathbf{q})\mathbf{F}_{\text{contact}}, \quad (5.1)$$

where:

- $M(\mathbf{q}) \in \mathbb{R}^{n \times n}$  is the mass matrix (from segment inertial properties)
- $C(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{n \times n}$  is the Coriolis/centrifugal matrix
- $g(\mathbf{q}) \in \mathbb{R}^n$  is the gravity vector
- $R(\mathbf{q}) \in \mathbb{R}^{n \times p}$  is the muscle moment arm matrix ( $p$  muscles,  $n$  DOF)
- $\mathbf{F}_{\text{muscle}} \in \mathbb{R}^p$  are the muscle forces (from Hill models, Chapter 3)
- $J_c(\mathbf{q})$  is the contact Jacobian
- $\mathbf{F}_{\text{contact}}$  are external contact forces

The key difference from Volume I's  $\boldsymbol{\tau} = B\mathbf{u}$  is that joint torques are no longer direct control inputs. Instead:

$$\boldsymbol{\tau}_{\text{net}} = R(\mathbf{q})\mathbf{F}_{\text{muscle}} = \sum_{i=1}^p \mathbf{r}_i(\mathbf{q})F_{\text{muscle},i}, \quad (5.2)$$

where  $\mathbf{r}_i(\mathbf{q}) \in \mathbb{R}^n$  is the  $i$ -th column of  $R$ , giving the moment arm vector of muscle  $i$  across all joints. This moment-arm mapping is the standard bridge between muscle models and generalized coordinates in musculoskeletal simulation [33, 7].

### 5.1.2 Bi-Articular Muscles

Unlike engineering actuators that span a single joint, many muscles cross two or more joints. These **bi-articular** (or multi-articular) muscles create mechanical coupling between joints that is not captured by the simple one-actuator-per-joint abstraction common in introductory robotics.

#### Biomechanical Principle 5.1. The Coordination Function of Bi-articular Muscles

Bi-articular muscles serve two distinct functions:

1. **Energy transfer:** they can transport mechanical energy from one joint to another without requiring a separate distal actuator. When the proximal joint extends (muscle shortening at the proximal end) and the distal joint flexes (muscle lengthening at the distal end), mechanical power can be redistributed from proximal to distal coordinates [33, 20]. The metabolic cost still depends on activation, contraction velocity, and force history.
2. **Coordination constraint:** they link joint motions, reducing the effective degrees of freedom and simplifying the control problem.

This is a form of *mechanical intelligence*—the morphology of the musculoskeletal system performs part of the control task. The gastrocnemius, for example, crosses both the knee and ankle, coupling knee extension to ankle plantarflexion during the push-off phase of gait.

**Control-Theoretic Perspective:** The matrix  $R(\mathbf{q})$  can be read as a state-dependent input map from muscle forces to generalized torques. Bi-articular muscles may create dependencies among its columns, which is one reason the effective control directions in a musculoskeletal model can be lower-dimensional than the raw muscle count. Geometric control language is useful for describing that reduction [1], but the specific structure of  $R(\mathbf{q})$  remains anatomy- and model-dependent.

```

1 import numpy as np
2
3 def compute_joint_torques(
4     moment_arms: np.ndarray,
5     muscle_forces: np.ndarray
6 ) -> np.ndarray:
7     """Compute net joint torques from muscle forces and moment arms.
8
9     Parameters
10    -----
11    moment_arms : np.ndarray, shape (n_joints, n_muscles)
12        Moment arm matrix  $R(\mathbf{q})$ . Each column gives the moment arm
13        of one muscle across all joints. Sign convention: positive
14        moment arm produces positive joint torque (typically flexion).
15    muscle_forces : np.ndarray, shape (n_muscles,)
16        Force produced by each muscle (always positive).
17
18    Returns
19    -----
20    np.ndarray, shape (n_joints,)
21        Net joint torques.
22    """
23    return moment_arms @ muscle_forces
24
25
26 # Example: 2-DOF elbow system with 4 muscles
27 # Joints: shoulder flexion, elbow flexion
28 # Muscles: anterior deltoid, biceps (biarticular),
29 #          triceps (biarticular), brachialis
30 #
31 # Convention: R has shape (n_joints, n_muscles) so that
32 #  $\tau = R @ F$  maps the muscle-force vector to a joint-torque
33 # vector via a single matrix-vector product. Each row is a joint
34 # and each column is a muscle. This is the shape that
35 # compute_joint_torques() expects.
36 R = np.array([
37     # deltoid, biceps, triceps, brachialis
38     [0.05, 0.03, -0.025, 0.0], # row 0: shoulder moment arms (m)
39     [0.0, 0.04, -0.020, 0.03], # row 1: elbow moment arms (m)
40 ]) # Shape: (2 joints, 4 muscles) -- ready to use as-is
41
42 # Note: biceps has moment arms at BOTH joints (bi-articular)
43 muscle_forces = np.array([200.0, 150.0, 100.0, 120.0]) # Newtons
44
45 tau = compute_joint_torques(R, muscle_forces)

```

```

46 print(f"Shoulder torque: {tau[0]:.1f} N*m")
47 print(f"Elbow torque:      {tau[1]:.1f} N*m")

```

Listing 5.1: Computing joint torques from muscle forces through moment arms.

### 5.1.3 Co-Contraction and Joint Stiffness

When antagonist muscles co-contract (both agonist and antagonist are active simultaneously), the net joint torque may be zero but the joint **stiffness** increases. This is because each active muscle acts as a spring:

$$K_{\text{joint}} = \sum_{i=1}^p r_i^2(\mathbf{q}) \cdot \frac{\partial F_{\text{muscle},i}}{\partial \ell_i^M}, \quad (5.3)$$

where  $\partial F / \partial \ell^M$  is the short-range stiffness of the muscle, which depends on activation level.

Co-contraction is metabolically expensive but biomechanically important: it allows the nervous system to modulate joint *impedance* independently of net joint *torque*. This is better understood as a biologically compliant analogue of impedance control rather than a literal duplicate of a rigid torque source [33, 19].

Position-dependent stiffness modulation gives a useful control vocabulary for discussing state-dependent actuation. Geometric control theory describes how state-dependent inputs change closed-loop structure [1], but the actual stiffness law in biomechanics still has to come from muscle, tendon, and contact modeling rather than from control theory alone.

## 5.2 The Mass Matrix of the Human Body

Computing the mass matrix  $M(\mathbf{q})$  for a full musculoskeletal model follows the same recursive algorithms as Volume 0, Chapter 10 (the Composite Rigid Body Algorithm), using the segment inertial properties from Chapter 2. The key computational steps are:

1. Define the kinematic tree (segment connectivity and joint types)
2. Compute forward kinematics using the product of exponentials
3. Apply the CRBA to compute  $M(\mathbf{q})$

For a 20-DOF model (pelvis + 2 legs + trunk + 2 arms),  $M(\mathbf{q})$  is a  $20 \times 20$  symmetric positive-definite matrix.

## Exercises

1. **Bi-articular Energy Transfer.** For the gastrocnemius (crossing knee and ankle), show mathematically how knee extension can produce ankle torque through the bi-articular coupling.
2. **Co-Contraction Analysis.** Compute the joint stiffness when biceps and triceps co-contract at 50% activation with the elbow at  $90^\circ$ .

3. **Mass Matrix Properties.** For a 2-segment arm model, compute  $M(\mathbf{q})$  at three configurations and verify positive-definiteness. Plot the condition number as a function of elbow angle.
4. **(Programming.)** Implement the CRBA for a 3-segment leg model using the de Leva segment properties. Compare with Pinocchio output.
5. **(Programming.)** Simulate a 2-DOF arm with 4 muscles (as in the code listing) performing a reaching movement. Use prescribed muscle activations and compute the resulting trajectory via forward dynamics.

## Chapter 6

# Inverse Problems in Biomechanics

---

**Draft Notice.** This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

---

In biomechanics, we almost never measure forces directly. We measure motion and infer the forces that caused it.

---

### 6.1 The Inverse Dynamics Pipeline

Inverse dynamics is one of the central tools in biomechanics [29, 7]: given measured kinematics  $\mathbf{q}(t)$ ,  $\dot{\mathbf{q}}(t)$ ,  $\ddot{\mathbf{q}}(t)$  and external forces  $\mathbf{F}_{\text{ext}}(t)$ , compute the net joint torques. By itself, however, the calculation stops at net generalized loads; moving from those loads to individual muscle forces requires an additional redundancy-resolution or forward-simulation model. The complete pipeline, illustrated in Figure 6.1, transforms raw experimental data into progressively more model-dependent estimates:

$$\boldsymbol{\tau}(t) = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) - \mathbf{J}_c^T(\mathbf{q})\mathbf{F}_{\text{ext}}. \quad (6.1)$$

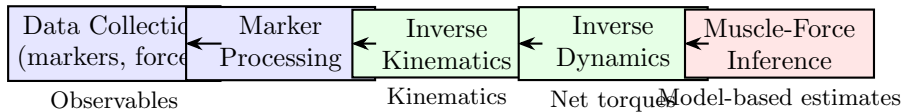


Figure 6.1: The Inverse Dynamics Pipeline: From Raw Experimental Data to Net Joint Moments and Subsequent Model-Based Muscle-Force Estimation.

The standard pipeline comprises five stages:

1. **Data Collection:** Motion capture markers + force plates
2. **Marker Processing:** Filtering, gap-filling, labeling

3. **Inverse Kinematics:** Fit skeleton model to marker data
4. **Inverse Dynamics:** Compute joint torques from RNEA
5. **Muscle Force Estimation:** Infer candidate muscle-force distributions using optimization or controller assumptions

Stages 1–4 are the standard inverse-kinematics/inverse-dynamics pipeline. Stage 5 is an additional modeling layer: it does not follow uniquely from the measured motion and external-force data, and different optimization or control assumptions can produce different muscle-force histories for the same net joint moments.

### 6.1.1 Inverse Kinematics

Given marker positions  $\mathbf{m}_i^{\text{meas}}(t) \in \mathbb{R}^3$  for  $i = 1, \dots, N_m$  markers, inverse kinematics solves:

$$\mathbf{q}^*(t) = \arg \min_{\mathbf{q}} \sum_{i=1}^{N_m} w_i \|\mathbf{m}_i^{\text{model}}(\mathbf{q}) - \mathbf{m}_i^{\text{meas}}(t)\|^2, \quad (6.2)$$

where  $\mathbf{m}_i^{\text{model}}(\mathbf{q})$  is the predicted marker position from the skeleton model and  $w_i$  are weighting factors.

```

1 import numpy as np
2 from scipy.optimize import minimize
3
4 def inverse_kinematics_frame(
5     marker_positions_measured: np.ndarray,
6     marker_positions_model_func,
7     q_initial: np.ndarray,
8     weights: np.ndarray | None = None
9 ) -> np.ndarray:
10     """Solve inverse kinematics for a single frame.
11
12     Parameters
13     -----
14     marker_positions_measured : np.ndarray, shape (n_markers, 3)
15         Measured marker positions.
16     marker_positions_model_func : callable
17         Function q -> (n_markers, 3) predicted marker positions.
18     q_initial : np.ndarray, shape (n_dof,)
19         Initial guess for joint angles.
20     weights : np.ndarray, shape (n_markers,), optional
21         Per-marker weights.
22
23     Returns
24     -----
25     np.ndarray, shape (n_dof,)
26         Optimal joint angles.
27     """
28     n_markers = marker_positions_measured.shape[0]
29     if weights is None:
30         weights = np.ones(n_markers)
31
32     def objective(q: np.ndarray) -> float:
33         m_model = marker_positions_model_func(q)
34         residuals = m_model - marker_positions_measured
35         return float(np.sum(weights[:, None] * residuals**2))

```

```

36
37     result = minimize(objective, q_initial, method='L-BFGS-B')
38     return result.x

```

Listing 6.1: Simplified inverse kinematics solver.

### 6.1.2 Bottom-Up vs. Top-Down Inverse Dynamics

**Bottom-up:** start from the ground contact (where forces are measured by force plates) and propagate upward through the kinematic chain. Most accurate when force plate data is available.

**Top-down:** start from the most distal segment and work inward. Useful when ground reaction forces are not available (e.g., swimming, throwing).

## 6.2 The Muscle Redundancy Problem

Inverse dynamics gives net joint torques  $\boldsymbol{\tau} \in \mathbb{R}^n$ , but we want individual muscle forces  $\mathbf{F} \in \mathbb{R}^p$  with  $p > n$ . The exact ratio between muscles and generalized coordinates depends on model granularity and how muscle groups are represented, but in all such cases the system is underdetermined:

$$R(\mathbf{q})\mathbf{F} = \boldsymbol{\tau}, \quad \mathbf{F} \geq 0. \quad (6.3)$$

From a modeling perspective, this is a constrained input-allocation problem. At a fixed configuration, the reachable torque set is the image of  $R(\mathbf{q})$  intersected with the nonnegative muscle-force constraints. When  $p > n$ , multiple muscle-force vectors can produce the same joint torque history, so additional physiological or optimization assumptions are needed to select among them.

### 6.2.1 Static Optimization

The most common approach minimizes a cost function:

$$\min_{\mathbf{F}} \sum_{i=1}^p \left( \frac{F_i}{F_{0,i}} \right)^\alpha \quad \text{subject to} \quad R(\mathbf{q})\mathbf{F} = \boldsymbol{\tau}, \quad 0 \leq F_i \leq a_i \cdot f_L(\tilde{\ell}_i) \cdot F_{0,i}, \quad (6.4)$$

where  $\alpha = 2$  (minimize sum of squared activations) or  $\alpha = 3$  (minimize metabolic energy proxy). The constraint  $F_i \leq a_i \cdot f_L \cdot F_{0,i}$  enforces the force-length and activation constraints from Chapter 3.

This optimization is one way to regularize the nonuniqueness of (6.3). The objective does not arise uniquely from the equations of motion; it encodes modeling priorities such as effort minimization, smoothness, or physiological plausibility. Different choices of  $\alpha$ , reserve actuators, activation dynamics, or regularization can yield different force-sharing solutions while reproducing the same net torque history [33, 7].

### 6.2.2 Computed Muscle Control (CMC)

CMC [28] combines forward dynamics with feedback control. It uses a PD controller to track the measured kinematics, with the controller output being the desired muscle excitations:

$$\ddot{\mathbf{q}}_{\text{des}} = \ddot{\mathbf{q}}_{\text{meas}} + K_v(\dot{\mathbf{q}}_{\text{meas}} - \dot{\mathbf{q}}) + K_p(\mathbf{q}_{\text{meas}} - \mathbf{q}), \quad (6.5)$$

where  $K_v$  and  $K_p$  are velocity and position feedback gains.

CMC is a tracking algorithm rather than a direct measurement technique. It can produce dynamically consistent motion in a forward simulation, but its output depends on model fidelity, actuator limits, controller tuning, and how closely the measured kinematics can actually be realized by the muscle-driven model. General nonlinear-control results explain why feedback can reduce tracking error, yet they do not by themselves validate a particular muscle-force reconstruction [1, 28].

## 6.3 Parameter Estimation

Many biomechanical parameters (segment masses, muscle strengths, joint axis locations) are not directly measurable and must be estimated from motion data.

### 6.3.1 Bayesian Framework

The parameter estimation problem is naturally Bayesian:

$$p(\boldsymbol{\theta}|\mathbf{y}) \propto p(\mathbf{y}|\boldsymbol{\theta}) \cdot p(\boldsymbol{\theta}), \quad (6.6)$$

where  $\boldsymbol{\theta}$  are model parameters,  $\mathbf{y}$  are observations,  $p(\mathbf{y}|\boldsymbol{\theta})$  is the likelihood, and  $p(\boldsymbol{\theta})$  is the prior (from literature values and physical constraints).

## Exercises

1. **Inverse Dynamics.** For a 2-segment pendulum model of the arm, compute the shoulder and elbow torques required during a reaching movement given  $\mathbf{q}(t) = (\sin(\pi t/T), \pi t/(2T))$ .
2. **Muscle Redundancy.** For a single-DOF elbow with 3 muscles (biceps, brachialis, brachioradialis), solve the static optimization for a given elbow torque. Show that the solution depends on the cost function exponent  $\alpha$ .
3. **(Programming.)** Implement the static optimization for a 2-DOF, 6-muscle system. Compare  $\alpha = 2$  and  $\alpha = 3$  solutions.
4. **(Programming.)** Implement an inverse kinematics solver for a 3-segment arm model with 7 markers. Test with synthetic marker data.

# Chapter 7

## Experimental Methods

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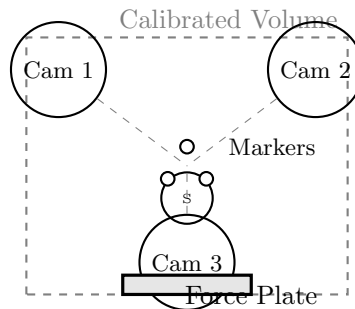
**Draft Notice.** This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

---

The quality of a biomechanical analysis is limited by the quality of the measurements, not the sophistication of the model.

---

### 7.1 Motion Capture Systems



#### 7.1.1 Marker-Based Systems

Optoelectronic motion capture systems (Vicon, Qualisys, OptiTrack) track retroreflective markers attached to the body surface. Standard configurations use 30–80 markers placed on bony landmarks following established protocols (Plug-in-Gait, Cleveland Clinic, ISB).

Key specifications for biomechanical applications [4]:

- **Sampling rate:** 100–500 Hz for human motion (higher for impacts)
- **Spatial accuracy:** < 1 mm in calibrated volume
- **Marker size:** 9–14 mm diameter
- **Latency:** important for real-time applications (< 10 ms achievable)

It is important to note that the instrument accuracy above refers to the camera system alone. The dominant error in marker-based biomechanics is not camera noise but *soft tissue artifact*—the relative motion between skin markers and the underlying bone—discussed in Section 7.1.3 below [17].

### 7.1.2 Markerless Systems

Computer vision approaches (OpenPose, MediaPipe, DeepLabCut) estimate joint positions from video without physical markers. Accuracy is currently 10–30 mm, insufficient for clinical-grade inverse dynamics but useful for field measurements and large-scale studies.

```

1 import numpy as np
2 from scipy.signal import butter, filtfilt
3
4 def process_mocap_data(
5     marker_data: np.ndarray,
6     sample_rate: float,
7     cutoff_freq: float = 6.0,
8     filter_order: int = 4
9 ) -> tuple[np.ndarray, np.ndarray, np.ndarray]:
10     """Process raw motion capture marker data.
11
12     Parameters
13     -----
14     marker_data : np.ndarray, shape (n_frames, n_markers, 3)
15         Raw marker positions in meters.
16     sample_rate : float
17         Sampling frequency in Hz.
18     cutoff_freq : float
19         Low-pass filter cutoff frequency in Hz.
20     filter_order : int
21         Butterworth filter order.
22
23     Returns
24     -----
25     tuple of np.ndarray
26         (positions, velocities, accelerations)
27         Each shape (n_frames, n_markers, 3).
28     """
29     # Design low-pass Butterworth filter
30     nyquist = sample_rate / 2.0
31     b, a = butter(filter_order, cutoff_freq / nyquist, btype='low')
32
33     n_frames, n_markers, _ = marker_data.shape
34     positions = np.zeros_like(marker_data)
35     velocities = np.zeros_like(marker_data)
36     accelerations = np.zeros_like(marker_data)
37
38     dt = 1.0 / sample_rate
39
40     for m in range(n_markers):
41         for axis in range(3):
42             # Filter position data
43             positions[:, m, axis] = filtfilt(
44                 b, a, marker_data[:, m, axis]
45             )
46
47     # Central difference for velocities
48     velocities[1:-1] = (positions[2:] - positions[:-2]) / (2 * dt)

```

```

49 velocities[0] = velocities[1]
50 velocities[-1] = velocities[-2]
51
52 # Central difference for accelerations
53 accelerations[1:-1] = (
54     positions[2:] - 2 * positions[1:-1] + positions[:-2]
55 ) / dt**2
56 accelerations[0] = accelerations[1]
57 accelerations[-1] = accelerations[-2]
58
59 return positions, velocities, accelerations

```

Listing 7.1: Motion capture data processing pipeline.

### 7.1.3 Soft Tissue Artifact

Soft tissue artifact (STA) is the relative motion between skin-mounted markers and the underlying bone caused by deformation of skin, subcutaneous fat, and muscle during movement. STA is widely recognized as the dominant source of error in marker-based motion capture, frequently exceeding the nominal camera resolution by more than an order of magnitude [4, 25].

The magnitude of STA varies strongly with marker location, anatomical region, task, and subject anthropometry. Representative values for the lower limb are 10–30 mm of skin-marker displacement relative to bone during walking and running, with the thigh segment the worst offender and the shank the best [4, 25]. STA is not random noise: it is task-correlated and heavily dependent on muscle contraction state, which means low-pass filtering cannot remove it.

Practical mitigation strategies include:

- **Rigid marker clusters** mounted on lightweight shells, which average out local skin motion over a neighborhood of markers.
- **Optimal pose estimation**, in which the rigid body pose of a segment is obtained by least-squares fitting of a cluster of markers rather than by differencing pairs of landmarks [4].
- **Global optimization / inverse kinematics** with joint constraints, which enforces an articulated model and redistributes STA across the chain.
- **Biplanar fluoroscopy** or bone pins when the research question demands sub-millimeter bone-to-bone accuracy.

STA should always be reported as part of the uncertainty budget of any kinetic analysis; inverse dynamics is sensitive to segment orientation errors, and moderate STA can produce large errors in computed joint moments.

### 7.1.4 Low-Pass Filter Cutoff Selection

Choosing the cutoff frequency for the low-pass Butterworth filter in `process_mocap_data()` is not a matter of taste: a cutoff that is too low attenuates the true signal (especially accelerations, whose energy extends to higher frequencies than positions), while a cutoff that is too high leaves noise that is amplified by numerical differentiation.

The standard objective procedure is **residual analysis**, due to Winter [29]. For a range of candidate cutoffs  $f_c$ , one computes the root-mean-square residual

$$R(f_c) = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i(f_c))^2},$$

where  $x_i$  is the raw signal and  $\hat{x}_i(f_c)$  is the filtered signal. Plotting  $R(f_c)$  vs.  $f_c$  produces a curve that is approximately linear in the high- $f_c$  regime (residuals reflect only noise) and rises sharply at low  $f_c$  (residuals include signal). The optimal cutoff is the intersection of the extrapolated linear noise-only segment with the  $R$ -axis, typically in the range 4–10 Hz for kinematics of whole-body locomotion and 10–20 Hz for running and impact-dominated tasks [29]. Higher cutoffs are required when downstream analysis needs accelerations or joint powers, since differentiation acts as a high-pass amplifier.

## 7.2 Electromyography (EMG)

EMG measures the electrical activity of muscles, providing a window into neural commands. Surface EMG (sEMG) is non-invasive; intramuscular EMG (using fine-wire or needle electrodes) targets deep muscles.

### 7.2.1 EMG Processing

The raw EMG signal must be processed before use:

1. **Band-pass filter:** 20–450 Hz (remove movement artifact and noise)
2. **Full-wave rectification:**  $|e(t)|$
3. **Envelope extraction:** low-pass filter at 3–6 Hz
4. **Normalization:** divide by maximum voluntary contraction (MVC) value

## 7.3 Force Plates

Force plates measure ground reaction forces (GRF) and moments. Standard specifications:

- 6-component measurement:  $F_x, F_y, F_z, M_x, M_y, M_z$
- Sampling rate: 1000–2000 Hz
- Range: 0–5000 N vertical, 0–2500 N horizontal
- Center of pressure (CoP) accuracy:  $< 2$  mm

The center of pressure is computed from force plate outputs:

$$x_{\text{CoP}} = -\frac{M_y + F_x \cdot d_z}{F_z}, \quad y_{\text{CoP}} = \frac{M_x - F_y \cdot d_z}{F_z}, \quad (7.1)$$

where  $d_z$  is the distance from the force plate surface to the transducer plane.

## 7.4 Inertial Measurement Units (IMUs)

IMUs combine accelerometers, gyroscopes, and magnetometers in a compact package. They provide:

- Angular velocity:  $\boldsymbol{\omega}(t) \in \mathbb{R}^3$
- Linear acceleration:  $\boldsymbol{a}(t) \in \mathbb{R}^3$
- Magnetic field:  $\boldsymbol{B}(t) \in \mathbb{R}^3$  (for heading)

Orientation estimation from IMU data is a state estimation problem solvable using the extended Kalman filter (Volume I, Chapter 6) or complementary filters.

## 7.5 Data Synchronization

Multiple measurement systems must be temporally synchronized. Best practices:

- Hardware sync: trigger all systems from a common clock pulse
- Minimum: force plate and motion capture on the same clock
- EMG-to-motion delay: typically 10–80 ms (electromechanical delay)
- Cross-correlation for post-hoc alignment when hardware sync is unavailable

## Exercises

1. **Filter Selection.** Using the provided code, filter a synthetic marker trajectory with cutoff frequencies of 4, 6, 8, and 12 Hz. Plot the effect on position, velocity, and acceleration. Discuss the trade-off between noise reduction and signal distortion.
2. **CoP Computation.** Given force plate data from a vertical jump ( $F_z$  peak = 3000 N), compute the center of pressure trajectory during takeoff.
3. **EMG Processing.** Process a synthetic EMG signal (sum of sinusoids + white noise) through the full processing pipeline. Plot each stage.
4. **(Programming.)** Implement an IMU orientation estimator using a complementary filter. Test with synthetic gyroscope and accelerometer data.

# Chapter 8

## Inference on Biological Systems

---

**Draft Notice.** This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

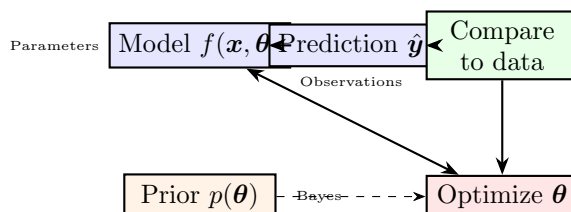
---

We cannot open the body and read the parameters. We must infer them from the traces left by motion.

---

### 8.1 Parameter Identification From Motion Data

Biological system parameters (muscle strengths, tendon slack lengths, segment inertias) cannot typically be measured directly in vivo. They must be inferred from observable motion data using optimization and estimation techniques.



#### 8.1.1 Maximum Likelihood Estimation

Given a parameterized model  $\dot{\mathbf{x}} = f(\mathbf{x}, \boldsymbol{\theta})$  with parameters  $\boldsymbol{\theta}$  and noisy observations  $\mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k$  at times  $t_k$ , the maximum likelihood estimate is:

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \sum_{k=1}^N \|\mathbf{y}_k - h(\mathbf{x}_k(\boldsymbol{\theta}))\|_{R^{-1}}^2, \quad (8.1)$$

where  $R$  is the measurement noise covariance.

### 8.1.2 Bayesian Estimation With Physical Constraints

The Bayesian framework (introduced in Chapter 6) is particularly valuable for biomechanical parameter estimation because it naturally incorporates prior knowledge from the literature:

```

1 import numpy as np
2 from scipy.optimize import minimize
3
4 def log_posterior(
5     theta: np.ndarray,
6     observations: np.ndarray,
7     forward_model,
8     prior_mean: np.ndarray,
9     prior_cov: np.ndarray,
10    noise_cov: np.ndarray
11 ) -> float:
12     """Compute the log-posterior for parameter estimation.
13
14     Parameters
15     -----
16     theta : np.ndarray
17         Model parameters.
18     observations : np.ndarray, shape (n_obs, n_dims)
19         Observed data.
20     forward_model : callable
21         theta -> predicted_observations.
22     prior_mean : np.ndarray
23         Prior parameter mean.
24     prior_cov : np.ndarray
25         Prior parameter covariance.
26     noise_cov : np.ndarray
27         Observation noise covariance.
28
29     Returns
30     -----
31     float
32         Negative log-posterior (for minimization).
33     """
34     # Log-likelihood
35     predicted = forward_model(theta)
36     residuals = observations - predicted
37     R_inv = np.linalg.inv(noise_cov)
38     log_likelihood = -0.5 * np.sum(residuals @ R_inv * residuals)
39
40     # Log-prior
41     prior_residuals = theta - prior_mean
42     P_inv = np.linalg.inv(prior_cov)
43     log_prior = -0.5 * prior_residuals @ P_inv @ prior_residuals
44
45     return -(log_likelihood + log_prior)
46
47
48 def map_estimate(
49     observations: np.ndarray,
50     forward_model,
51     prior_mean: np.ndarray,
52     prior_cov: np.ndarray,
53     noise_cov: np.ndarray,
54     bounds: list[tuple[float, float]] | None = None
55 ) -> np.ndarray:
56     """Compute Maximum A Posteriori (MAP) parameter estimate.
```

```

57
58     Parameters
59     -----
60     observations : np.ndarray
61         Observed data.
62     forward_model : callable
63         Parameter-to-observation mapping.
64     prior_mean : np.ndarray
65         Prior mean from literature.
66     prior_cov : np.ndarray
67         Prior uncertainty.
68     noise_cov : np.ndarray
69         Measurement noise covariance.
70     bounds : list of tuple, optional
71         Physical bounds on parameters.
72
73     Returns
74     -----
75     np.ndarray
76         MAP parameter estimate.
77     """
78     result = minimize(
79         log_posterior,
80         prior_mean,
81         args=(observations, forward_model, prior_mean,
82             prior_cov, noise_cov),
83         method='L-BFGS-B',
84         bounds=bounds
85     )
86     return result.x

```

Listing 8.1: Bayesian parameter estimation for musculoskeletal model.

## 8.2 System Identification for Biological Systems

System identification techniques from control theory (Volume I) apply to biomechanical systems with modifications:

1. **Joint impedance identification:** Perturb the limb and measure the response. The stiffness, damping, and inertia of the joint can be estimated from the transfer function.
2. **Muscle parameter identification:** Optimize muscle parameters ( $F_0$ , optimal length, tendon slack length) to match experimental force or torque data.
3. **Neural control identification:** Estimate the feedback gains used by the nervous system from perturbation experiments.

## 8.3 Machine Learning Approaches

Modern ML approaches to biomechanical inference include:

- Physics-informed neural networks (PINNs) that enforce equations of motion
- Gaussian process regression for joint angle estimation from sparse data
- Deep learning for markerless motion capture pose estimation

## Exercises

1. **MAP Estimation.** Using the provided code, estimate the optimal fiber length of a biceps muscle from simulated isometric force data at 5 joint angles.
2. **Identifiability Analysis.** For a Hill-type muscle model, analyze which parameters can be uniquely estimated from isometric force-angle data alone. Which require dynamic data?
3. **(Programming.)** Implement a joint impedance identification experiment: simulate a perturbed pendulum and estimate stiffness and damping from the response.

## Chapter 9

# Deformable Bodies and Continuum Mechanics

---

**Draft Notice.** This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

---

When the rigid-body assumption fails, we do not abandon mechanics. We embrace the continuum.

---

### 9.1 When Rigid-Body Assumptions Fail

The rigid-body model fails when:

1. **Internal stresses are needed:** to predict injury, fracture, or tissue damage
2. **Deformation affects kinematics:** shaft flexibility in golf, spine compliance
3. **Wave propagation matters:** impact biomechanics, blast injury
4. **Fluid-structure interaction:** blood flow, synovial fluid, respiratory mechanics

### 9.2 Beam Models for Long Bones and Shafts

When deformation is small and the body is slender (length  $\gg$  diameter), beam theory provides an efficient model. A schematic representation of the finite element beam model is shown in Figure 9.1.

For a Euler-Bernoulli beam:

$$EI \frac{\partial^4 w}{\partial x^4} + \rho A \frac{\partial^2 w}{\partial t^2} = f(x, t), \quad (9.1)$$

where  $E$  is the elastic modulus,  $I$  is the second moment of area,  $\rho$  is density,  $A$  is cross-sectional area,  $w(x, t)$  is transverse displacement, and  $f$  is the distributed load.

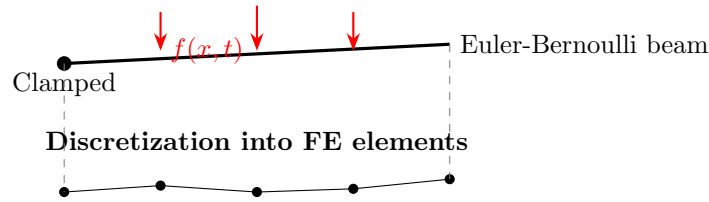


Figure 9.1: Finite Element Beam Model: Continuous Euler-Bernoulli Beam (Top) and Its Discretization Into Nodal Elements (Bottom).

### 9.2.1 Modal Analysis

The solution is expanded in modes:

$$w(x, t) = \sum_{k=1}^N \phi_k(x) \eta_k(t), \quad (9.2)$$

where  $\phi_k(x)$  are mode shapes and  $\eta_k(t)$  are modal coordinates. This converts the PDE to a system of ODEs that can be appended to the rigid-body equations of motion.

### 9.2.2 Golf Club Shaft Flexibility

The golf club shaft is the most accessible example of a flexible link in sport biomechanics. A typical shaft has:

- Length:  $\sim 1.1$  m
- First bending mode: 4–6 Hz (driver), 6–10 Hz (irons)
- Tip deflection at impact: 5–15 cm
- Lead/lag deflection: 2–5 cm
- Toe-down droop: 3–8 cm (gravity effect during downswing)

```

1 import numpy as np
2 from scipy.linalg import eigh
3
4 def shaft_modal_analysis(
5     n_elements: int = 20,
6     length: float = 1.1,
7     EI_root: float = 50.0,
8     EI_tip: float = 15.0,
9     rho_A: float = 0.15
10 ) -> tuple[np.ndarray, np.ndarray]:
11     """Compute natural frequencies and mode shapes of a tapered shaft.
12
13     Uses finite element beam model with linear EI taper.
14
15     Parameters
16     -----
17     n_elements : int
18         Number of finite elements.

```

```

19     length : float
20         Shaft length in meters.
21     EI_root : float
22         Flexural rigidity at root (grip end) in N*m^2.
23     EI_tip : float
24         Flexural rigidity at tip (head end) in N*m^2.
25     rho_A : float
26         Mass per unit length in kg/m.
27
28     Returns
29     -----
30     tuple of np.ndarray
31         (frequencies_hz, mode_shapes)
32     """
33     n_nodes = n_elements + 1
34     n_dof = 2 * n_nodes # displacement + rotation at each node
35     h = length / n_elements
36
37     K = np.zeros((n_dof, n_dof))
38     M = np.zeros((n_dof, n_dof))
39
40     for e in range(n_elements):
41         # Local EI for this element (linear taper)
42         xi = (e + 0.5) / n_elements
43         EI_local = EI_root + (EI_tip - EI_root) * xi
44
45         # Euler-Bernoulli beam element stiffness matrix
46         k_e = EI_local / h**3 * np.array([
47             [12, 6*h, -12, 6*h],
48             [6*h, 4*h**2, -6*h, 2*h**2],
49             [-12, -6*h, 12, -6*h],
50             [6*h, 2*h**2, -6*h, 4*h**2]
51         ])
52
53         # Consistent mass matrix
54         m_e = rho_A * h / 420 * np.array([
55             [156, 22*h, 54, -13*h],
56             [22*h, 4*h**2, 13*h, -3*h**2],
57             [54, 13*h, 156, -22*h],
58             [-13*h, -3*h**2, -22*h, 4*h**2]
59         ])
60
61         # Assembly
62         dofs = [2*e, 2*e+1, 2*e+2, 2*e+3]
63         for i_idx, i in enumerate(dofs):
64             for j_idx, j in enumerate(dofs):
65                 K[i, j] += k_e[i_idx, j_idx]
66                 M[i, j] += m_e[i_idx, j_idx]
67
68     # Boundary conditions: clamped at root (grip)
69     free_dofs = list(range(2, n_dof)) # Remove first node
70     K_free = K[np.ix_(free_dofs, free_dofs)]
71     M_free = M[np.ix_(free_dofs, free_dofs)]
72
73     # Solve eigenvalue problem
74     eigenvalues, eigenvectors = eigh(K_free, M_free)
75
76     # Convert to Hz
77     frequencies = np.sqrt(np.abs(eigenvalues)) / (2 * np.pi)
78

```

```

79     return frequencies[:6], eigenvectors[:, :6]
80
81
82 # Example
83 freqs, modes = shaft_modal_analysis()
84 print("Natural frequencies (Hz):")
85 for i, f in enumerate(freqs):
86     print(f"  Mode {i+1}: {f:.1f} Hz")

```

Listing 9.1: Modal analysis of a golf club shaft.

### 9.3 Finite Element Methods (Overview)

For complex geometries (vertebral bodies, skull, articular surfaces), the finite element method (FEM) provides the most general approach. While a full treatment is beyond this text's scope, the key concept is:

The continuous domain  $\Omega$  is divided into elements, and the displacement field is approximated using shape functions:

$$\mathbf{u}(\mathbf{x}, t) \approx \sum_i N_i(\mathbf{x}) \mathbf{u}_i(t), \quad (9.3)$$

leading to the assembled system:

$$M_{\text{FEM}} \ddot{\mathbf{u}} + C_{\text{FEM}} \dot{\mathbf{u}} + K_{\text{FEM}} \mathbf{u} = \mathbf{f}_{\text{ext}}. \quad (9.4)$$

### Exercises

1. **Shaft Frequency.** Using the provided code, compute how the first natural frequency changes as the tip-to-root EI ratio varies from 0.1 to 0.9. Plot the result.
2. **Mode Shapes.** Plot the first three mode shapes of the shaft. Identify nodes (zero-crossings) and antinodes (maximum displacement).
3. **Timescale Separation.** If the golf downswing takes 0.25 s, estimate the number of oscillation cycles for each of the first 3 modes. Which modes are quasi-static?
4. **(Programming.)** Add a tip mass (clubhead, 0.2 kg) to the shaft FEM model and recompute the natural frequencies. How does the clubhead mass affect the frequencies?

## Chapter 10

# Applications of Control Theory to Biological Systems

---

**Draft Notice.** This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

---

The body is not a robot that happens to be made of meat. It is a system that has evolved to exploit its own physics.

---

### 10.1 Forward Dynamics Simulation

Forward dynamics simulation integrates the equations of motion given muscle excitations  $\mathbf{u}(t)$ . A feedback control loop for biological systems is illustrated in Figure 10.1.

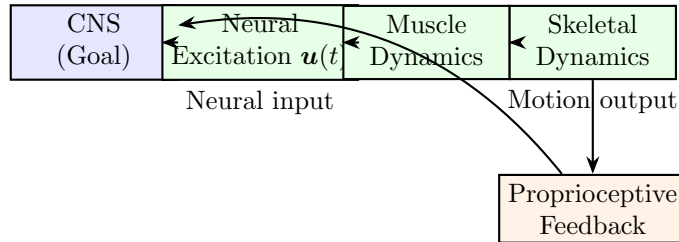


Figure 10.1: Biological Feedback Control Loop: CNS Goal Signals Drive Neural Excitations Through Muscle and Skeletal Dynamics, With Proprioceptive Feedback Closing the Loop.

$$\ddot{\mathbf{q}} = M(\mathbf{q})^{-1} [R(\mathbf{q})\mathbf{F}_{\text{muscle}} + J_c^T \mathbf{F}_c - C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - g(\mathbf{q})], \quad (10.1)$$

coupled with the muscle state equations (Chapter 3):

$$\dot{a}_i = (u_i - a_i)/\tau_i, \quad (10.2)$$

$$\dot{\ell}_i^M = v_i^M(a_i, \ell_i^M, \ell_i^{MT}(\mathbf{q})). \quad (10.3)$$

```

1 import numpy as np
2 from scipy.integrate import solve_ivp
3 from typing import Callable
4
5 def forward_dynamics_simulation(
6     mass_matrix_func: Callable,
7     coriolis_func: Callable,
8     gravity_func: Callable,
9     moment_arm_func: Callable,
10    muscle_force_func: Callable,
11    excitation_func: Callable,
12    q0: np.ndarray,
13    qdot0: np.ndarray,
14    a0: np.ndarray,
15    t_span: tuple[float, float],
16    n_eval: int = 500
17 ) -> dict:
18     """Simulate forward dynamics of a musculoskeletal system.
19
20     Parameters
21     -----
22     mass_matrix_func : callable
23         q -> M(q), shape (n_dof, n_dof)
24     coriolis_func : callable
25         (q, qdot) -> C(q, qdot) @ qdot, shape (n_dof,)
26     gravity_func : callable
27         q -> g(q), shape (n_dof,)
28     moment_arm_func : callable
29         q -> R(q), shape (n_dof, n_muscles)
30     muscle_force_func : callable
31         (a, l_M, v_M) -> F_muscle for each muscle
32     excitation_func : callable
33         t -> u(t), shape (n_muscles,)
34     q0 : np.ndarray
35         Initial joint angles.
36     qdot0 : np.ndarray
37         Initial joint velocities.
38     a0 : np.ndarray
39         Initial muscle activations.
40     t_span : tuple
41         (t_start, t_end)
42     n_eval : int
43         Number of output time points.
44
45     Returns
46     -----
47     dict
48         Keys: 't', 'q', 'qdot', 'a', 'tau'
49     """
50     n_dof = len(q0)
51     n_muscles = len(a0)
52     t_eval = np.linspace(t_span[0], t_span[1], n_eval)
53
54     def dynamics(t: float, state: np.ndarray) -> np.ndarray:
55         q = state[:n_dof]
56         qdot = state[n_dof:2*n_dof]
57         a = np.clip(state[2*n_dof:], 0.0, 1.0)
58
59         # Muscle forces (simplified: assume known fiber length)
60         F_muscle = np.array([

```

```

61         a[i] * 500.0 # Simplified: F = a * F0
62         for i in range(n_muscles)
63     ])
64
65     # Joint torques
66     R = moment_arm_func(q)
67     tau = R @ F_muscle
68
69     # Skeletal dynamics
70     M = mass_matrix_func(q)
71     C_qdot = coriolis_func(q, qdot)
72     g = gravity_func(q)
73     qddot = np.linalg.solve(M, tau - C_qdot - g)
74
75     # Activation dynamics
76     u = excitation_func(t)
77     tau_act = 0.015 # Illustrative placeholder; replace with a cited
model value.
78     tau_deact = 0.050 # Illustrative placeholder; replace with a cited
model value.
79     adot = np.where(
80         u > a,
81         (u - a) / tau_act,
82         (u - a) / tau_deact
83     )
84
85     return np.concatenate([qdot, qddot, adot])
86
87     state0 = np.concatenate([q0, qdot0, a0])
88     sol = solve_ivp(dynamics, t_span, state0, t_eval=t_eval,
89                     method='RK45', max_step=0.001)
90
91     return {
92         't': sol.t,
93         'q': sol.y[:n_dof].T,
94         'qdot': sol.y[n_dof:2*n_dof].T,
95         'a': sol.y[2*n_dof:].T,
96     }

```

Listing 10.1: Forward dynamics integration for a musculoskeletal model.

## 10.2 Predictive Simulation

Predictive simulation uses trajectory optimization (Volume I, Chapter 5) to find the muscle excitations that produce a desired motion while minimizing a cost:

$$\min_{\mathbf{u}(\cdot)} \int_0^T \mathcal{L}(\mathbf{x}, \mathbf{u}) dt \quad \text{s.t.} \quad \dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}), \mathbf{x}(0) = \mathbf{x}_0, h(\mathbf{x}, \mathbf{u}) \leq 0, \quad (10.4)$$

where  $\mathcal{L}$  is a physiologically motivated cost (e.g., metabolic energy, muscle fatigue, deviation from desired kinematics) and  $h$  captures physical constraints (joint limits, muscle force capacity, contact constraints).

Agrachev & Sachkov [1], Theorem 8.1 (Pontryagin's Maximum Principle), establishes the fundamental existence and optimality conditions for such optimal control problems. Their treatment covers affine systems with constraints and provides both theoretical guarantees and computational methods (the method of characteristics) for solving OCPs of this form.

## 10.3 Prosthetics and Exoskeleton Control

The control theory of Volumes I–II applies directly to assistive devices:

- **Powered prostheses:** trajectory tracking with impedance modulation
- **Exoskeletons:** assist-as-needed control using contraction theory
- **FES (Functional Electrical Stimulation):** direct muscle activation control for paralyzed limbs

## 10.4 Sport Biomechanics Applications

### 10.4.1 The Golf Swing as a Biomechanical System

The golf swing demonstrates every concept in this volume:

1. Multi-segment kinematic chain (15+ DOF)
2. Muscle-driven actuation with co-contraction
3. Bi-articular coupling (forearm muscles, lats)
4. Passive dynamics exploitation (wrist release)
5. Shaft flexibility (deformable body)
6. Ground reaction force coupling (closed chain at the feet)

The complete biomechanical analysis integrates:

- Motion capture (kinematics)
- Force plates (ground reaction forces)
- EMG (muscle activation patterns)
- Inverse dynamics (joint torques)
- Static optimization (muscle forces)
- Forward dynamics validation (predictive simulation)

## 10.5 Synthesis: From Theory to Application

The progression through this volume mirrors the progression through the series:

| Volume  | Concept            | Biomechanical Application          | Agrachev & Sachkov |
|---------|--------------------|------------------------------------|--------------------|
| Vol 0   | Screw axes, PoE    | Joint kinematics, ISA analysis     |                    |
| Vol I   | Contraction theory | Stability of muscle-driven systems |                    |
| Vol I   | DDP/iLQR           | Optimal muscle excitation patterns |                    |
| Vol II  | Trajectory tubes   | Motor variability envelopes        |                    |
| Vol II  | Phase variables    | Gait cycle parameterization        |                    |
| Vol III | Hill model         | Force prediction                   |                    |
| Vol III | Inverse dynamics   | Joint torque estimation            |                    |
| Vol IV  | Motor control      | Neural command generation          |                    |

## Exercises

1. **Forward Simulation.** Using the provided code (adapted for a simple 2-DOF model), simulate a reaching movement with prescribed muscle excitation ramps.
2. **Predictive vs. Tracking.** Compare the muscle activations from static optimization (tracking measured kinematics) with those from predictive simulation (optimizing a metabolic cost).
3. **Prosthetic Control.** Design an impedance controller for a prosthetic knee that provides different stiffness during stance and swing phases.
4. **(Programming.)** Implement a complete inverse dynamics  $\rightarrow$  static optimization pipeline for a 3-DOF planar arm model with 6 muscles.

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